P∞-F: Primes Against Sonin Space — the (B2b) Assault and the Anatomy of Window Weil Positivity

Leonardo Murillo Montero
Independent researcher

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Draft 2 — 2026-06-12 (referee pass; launched 2026-06-11). Sequel to P∞-E §§19–26. The original target: **(B2b)** *soft semilocal Sonin positivity*; the arc that followed (§§7–16: theorems F.1–F.18, ~20 certificates, five fired-and-diagnosed falsifier batteries) ended somewhere sharper. Method: GPD (Paper 35) — bold mechanism, pre-registered falsifiers, exact-arithmetic settlement, **beliefs disconfirmed and replaced where the data demands; all corpses retained in place.**

State of the ledger (the referee's map — read this first). Each section's closing "ledger" records the *state of belief at that strike*, kept per GPD; the sequence of reductions, with their fates:

$$\text{(B2b)} \xrightarrow{\S7} \text{F.4.D} + \text{F.5}' \xrightarrow{\S8\text{–}9} \text{F.5}'' \xrightarrow{\S10} \text{merged (F.11)} \xrightarrow{\S11} (\mathrm{QS}_{\exp}) + \text{(a)}$$
$$\xrightarrow{\S12} \text{(H1) discharged} \xrightarrow{\S13} \boxed{\text{withdrawn}} \xrightarrow{\S14} (\mathrm{H0}')$$

Final form (§§14–16): RH $\iff$ (H0′) — window Weil positivity surviving the limit, E.20's hidden hypothesis — through the weld-amplifier, whose scaffolding this paper built and certified: the sampling-sector anatomy (§§7–9), the merged identity (§10), the count facet unconditional (§11), the Suzuki weld and the (H1) discharge (§12), the displacement calculus (§13), the literature coordinates of $(\mathrm{QS}_{\exp})$ (§§14–15), and the bookkeeping lemmas (§16). The two failure faces of (H0′) — zero location and zero degeneracy — are welded into one object family (F.18).

The post-consolidation strikes (§§17–22, added after the Draft-2 pass; the map continues):

$$(\mathrm{H0}') \xrightarrow{\S17} \text{three faces of } \epsilon_\lambda + (\mathrm{H0}'\text{-tail}) \xrightarrow{\S18} \text{horizon anatomy (F.19)} \xrightarrow{\S19} \text{innovation chain (F.21)}$$
$$\xrightarrow{\S20} \text{energy ledger (F.22)} \xrightarrow{\S21} R \approx Ce^{-0.342\lambda^2} \xrightarrow{\S22} \text{two-branch lifecycle, cumulative coordinates}$$

— the limit eliminated from the statement (F.21's center-out martingale immunity), the open step as wave-stationarity of two parity-branch profiles, with §22 rewritten in invariant coordinates after the invariance audit (per-rung spectral quantities of near-degenerate forms are not program-grade observables; cumulative/spectral-function coordinates are — F64-5).

Superseded intermediate claims are flagged where they stand (§6, §8.4, §11.3–4, §13.3, §21.1). Numbering note: the labels F.9 and F.20 are unassigned (kept for cross-reference stability; §18 ends at F.19, §19 opens at F.21). Certificates: `empirical/p_infinity/probe_pinfF_*.py` (twenty probes, run logs adjacent).

1. The foundation: the semilocal form is a multiplier (Theorem F.1)

Theorem F.1 [P; **certified to** 3×10^{-3}]**.** For f supported in $[\lambda^{-1}, \lambda]$,

$$QW_\lambda(f) \;=\; \frac{1}{\pi}\int_{\mathbb{R}} |\hat f(t)|^2\, W_\lambda(t)\, dt \;+\; 2\,\Re\big(\hat f(\tfrac{i}{2})\,\overline{\hat f(-\tfrac{i}{2})}\big), \qquad W_\lambda(t) := \theta'(t) - \sum_{2\le n\le\lambda^2} \frac{\Lambda(n)}{\sqrt{n}}\cos(t\log n).$$

Proof. Wiener–Khinchin on the multiplicative group: $(f^* * f)(n) + (f^* * f)(n^{-1}) = \frac{1}{\pi}\int |\widehat{f}|^2 \cos(t \log n)\, dt$; the arch term is already the θ'-multiplier (CCM eq. quadratsemi); the pole term is rank two. □

W_λ is the **λ-truncated zero density** (times π): formally $W_\infty(t) = \pi \sum_\gamma \delta(t-\gamma)$, the Guinand–Weil density. So the semilocal Weil form *samples the spectral energy $|\widehat{f}|^2$ against the truncated zero-counting measure*, plus one hyperbolic pole cell. The truncation makes W_λ pointwise negative in wells (certified: $W_\lambda(0) = -3.18$ at $\lambda = 1.5$) — yet the form is positive off-$\mathcal{E}$ (measured floor 0.28–0.61). **(B2b) is therefore a one-dimensional weighted uncertainty principle**: entire functions of exponential type $\log\lambda$ (Paley–Wiener of the window) cannot concentrate $|\widehat{f}|^2$ in the Gibbs wells of the truncated density — *unless* they carry the arithmetic coherence of the $\mathcal{E}$-sector. P∞'s light-cone, the RS body (P∞-E §17), and the cone deficit (§18,§20) all reappear in this single formula: the cone is where W_λ's oscillations live; the $\mathcal{E}$-cells are the functions tuned to them.

2. Historic Fiction (the fenced conjecture, as registered)

Conjecture F.2 (incoherence principle — AS ORIGINALLY HELD). The $\mathcal{E}$-sector consists exactly of the *coherent states* of the prime shift semigroup $\{S_n\}$; off it, the prime part of the form behaves like a sum of incoherent operators with square-root cancellation: compressed norm $O(\log\lambda)$, against the coherent bound $\sim 4\lambda$. Then (B2b) = [arch+pole Sonin floor] − [$O(\log\lambda)$ perturbation] > 0.

3. Pre-registered falsifiers

K1: measure the off-$\mathcal{E}$ compressed norm of the prime part across $\lambda = 1.5$–3.0; *falsified if growth is polynomial in λ.* **K2**: the off-$\mathcal{E}$ minimum of the full form; *falsified if it decays exponentially.* **K3**: the off-$\mathcal{E}$ minimum of arch+pole alone; *the mechanism is "prime smallness" only if K1 survives; if K3 shows a robust floor while K1 fires, the mechanism is misalignment, not smallness.*

4. Settlement (exact matrices, $\lambda = 1.5, 2.0, 2.5, 3.0$; $N = 10$–18; cells = all $n \equiv 0\,(4)$ with defect < 0.05, count $1, 3, 5, 8$)

λ	off-min (full)	off-min (arch+pole)	off-norm (prime)	coherent 4λ
1.5	0.280	0.550	0.486	6
2.0	0.568	0.829	0.896	8
2.5	0.299	**0.834**	1.296	10
3.0	0.611	**0.827**	1.752	12

K1 FIRED. Off-$\mathcal{E}$ prime norm grows $\sim \lambda^{1.84}$ — polynomial, not logarithmic. **Conjecture F.2 is dead as formulated**; the incoherence suppression is real (factor ~ 7 below coherent at $\lambda = 3$) but does not flatten the growth. Recorded, retained, replaced.

K2 SURVIVES. The full off-$\mathcal{E}$ minimum stays $O(1)$: $0.28, 0.57, 0.30, 0.61$ — no decay trend (the bounce correlates with new primes entering at λ^2 thresholds); (B2b)'s content is consistent with a constant, far above the needed 1/poly.

K3 SURVIVES, AND IT IS THE DISCOVERY. The arch+pole compression min is **flat at** ≈ 0.83 $(0.55, 0.829, 0.834, 0.827)$ — a λ-independent constant emerging from three different windows with different dimensions. And the full minimum stays positive even where the prime norm (1.75) *exceeds* the arch floor (0.83): the prime part's extreme directions are **misaligned** with the arch-minimal directions. The naive triangle inequality fails at $\lambda \geq 2.5$; positivity persists anyway.

5. The corrected mechanism, and the two theorems it calls for

Conjecture F.3 (relative form-boundedness — the post-disconfirmation belief). Off the $\mathcal{E}$-sector, the prime part is bounded *relative to the archimedean form, not in norm*: there is $\delta > 0$ with

$$\Big|\sum_{n\le\lambda^2}\Lambda(n)n^{-1/2}\langle f|T(n)f\rangle\Big| \;\le\; (1-\delta)\Big[\frac{1}{\pi}\int|\hat f|^2\theta'\,dt + \text{pole}\Big] \qquad \text{for } f\perp\mathcal{E}\text{-sector.}$$

The prime multiplier's large values sit where θ' is large (high t, the cone interior), and its low-t wells are exploitable only by $\mathcal{E}$-coherent functions — *relative* boundedness is exactly what the measured misalignment expresses. In multiplier language: $W_\lambda(t) \ge \delta\,\theta'(t) - C$ fails pointwise, but holds *after averaging against any off-$\mathcal{E}$ Paley–Wiener energy density* — a weighted uncertainty statement.

Target Theorem F.4 (the flat archimedean floor). Off the $\mathcal{E}$-sector, the arch+pole form is bounded below by an absolute constant (≈ 0.83 measured). This is the semilocal-window version of CC-2021's Sonin positivity, *quantified*; its λ-flatness strongly suggests a clean proof — plausibly: the off-$\mathcal{E}$ sector embeds into Sonin space up to exp-small angles (the $\mathcal{E}$-cells exhaust the near-$(P,\widehat{P})$-invariant directions, by Slepian theory), and on Sonin space the CC-2021 trace operator has an explicit spectral floor. **This is now the sharpest provable-looking statement on the board.**

Target Theorem F.5 (misalignment). F.3 for the prime part, proof ground: the prime multiplier $\sum\Lambda(n)n^{-1/2}\cos(t\log n)$ against off-$\mathcal{E}$ PW-densities — Montgomery–Vaughan-type mean-value technology (the multiplier is a Dirichlet polynomial in e^{it}-disguise; off-$\mathcal{E}$ energy densities have spectra avoiding the log-lattice; MV inequality bounds exactly such pairings). **The classical large-sieve/mean-value toolkit enters the architecture here** — for the first time with a precise, finite, falsifiable target.

6. Status and honesty *(historical — the ledger as of the launch strike; superseded per the preface map)*

Proven: F.1 [P]. Dead: F.2 (K1 fired; retained per GPD). Measured: the flat 0.83 floor; the misalignment; the $O(1)$ full minimum. Open: F.4 (Sonin-floor, the next strike), F.5 (MV-misalignment), and the bookkeeping stack of P∞-E §26. The ledger:

$$\text{RH} \;\Longleftarrow\; \text{(H1)} \;+\; \text{F.4} \;+\; \text{F.5} \;+\; \text{bookkeeping.}$$

Both F.4 and F.5 are statements about explicit objects with classical toolkits attached (Slepian angles; Montgomery–Vaughan), both measured true with margin, neither circular in RH. The assault continues there. *(Referee note: (H1) was discharged in §12; F.4/F.5 evolved per the preface map; the final form is §§14–16.)*

Anchors. P∞-E §§19–26 (the weld, the radical route, the sector split); CC 2021 (Sonin space; Thm 1); CCM 2511.22755 (quadratsemi); Wiener–Khinchin; Guinand–Weil; Slepian; Montgomery–Vaughan (mean values); Paper 35 (GPD); `probe_pinfF_b2b_falsifiers.py` (GREEN; K1 fired, K2/K3 survived).

7. The F.4 strike: the floor is the first zero — off-$\mathcal{E}$ is the sampling sector (Theorems F.4.A–C, Lemma F.4.B)

Certificate: `empirical/p_infinity/probe_pinfF_f4_floor.py` (GREEN; exact builds $\lambda \le 3$ reproduce §4's floors to 10^{-4}; the $\lambda = 3.5$ multiplier build cross-validated against the exact matrix at $\lambda = 3$ to 10^{-4}; the F.4.A inequality verified on 120 random draws $\times$ 3 bands). Predictions pre-registered in the probe header before running; one falsifier fired and is diagnosed below.

7.1 What was registered, and what the data did

Registered. P1: the floor is $2\theta'(\gamma_1) = 0.8105$ up to width corrections, λ-flat including a new window $\lambda = 3.5$. P2: the off-$\mathcal{E}$ minimizer *is* the γ_1 sampling representer; the second off-$\mathcal{E}$ eigenvalue sits at $2\theta'(\gamma_2) = 1.2076$

(zero-ladder), not at the band-edge alternative $2\theta'(17.58) = 1.0287$. P3: $\mathcal{E}$-coverage of band content *saturates at a band edge* $T^* \approx \gamma_1$.

Measured.

λ	cells	e_0 (floor)	e_1	e_2	$\bar{t}(v_0)$	$\lvert\langle v_0, \mathrm{rep}_1\rangle\rvert$	$\lvert\langle v_1, \mathrm{rep}_2\rangle\rvert$	pole share
1.5	1	0.5502	1.045	1.400	9.94	0.660	0.763	0.650
2.0	3	0.8293	1.113	1.341	14.54	0.945	0.865	0.003
2.5	5	0.8335	1.210	1.382	14.61	0.981	0.946	0.000
3.0	8	0.8266	1.206	1.357	14.48	0.978	1.000	0.015
3.5	11	0.8048	1.204	1.397	13.80	0.994	1.000	0.063

References: $2\theta'(\gamma_1) = 0.8105$, $2\theta'(\gamma_2) = 1.2076$, $2\theta'(\gamma_3) = 1.3814$. **P1 and P2 survive at full strength**: the floor converges onto $2\theta'(\gamma_1)$ as resolution refines; the off-$\mathcal{E}$ minimizer is the first zero's representer to overlap 0.994; the second and third off-$\mathcal{E}$ eigenvalues land on the zero ladder to three decimals ($e_1 = 1.206$ vs 1.2076; $\langle v_1, \mathrm{rep}_2\rangle = 1.000$), decisively rejecting the band-continuum alternative 1.029. The representers' own energies (EXP-B): $A(\mathrm{rep}_1) = 0.896, 0.877, 0.862, 0.859, 0.814$ across the sweep — the floor *is* the representer energy. The $\lambda = 1.5$ row is the under-resolved regime (window resolution $2\pi/L = 7.75$ exceeds zero spacings; the minimizer still has 61% sub-γ_1 mass and 65% pole share), exactly as the coverage-incomplete reading of §4's 0.55 predicted.

KF3 fired, and the diagnosis is criterion, not mathematics. The "full-cone" dictionary tier ($n \le 40$: ~ 20 images in a 16–19-dimensional even space) numerically saturates the whole space, so its zero residuals are span-vacuity; and the top band-Slepians are the *smoothest* band modes — the easiest to cover — so their small residuals against the cell span do not probe saturation. The informative geometry came from the representers themselves: rep_{γ_1}'s component inside the cell span is $0.001, 0.001, 0.000, 0.000, 0.012$ across the sweep — **the zero representers are orthogonal to the $\mathcal{E}$-sector** — and the dimension count closes: at $\lambda = 3$ (resolution 2.86), off-$\mathcal{E}$ dimension $11 = 10$ resolved zeros $+ 1$; at $\lambda = 3.5$, $11 = 10 + 1$ again (cutoff $\omega_N = 52.6$, $\gamma_{11} = 52.97$ just outside). **The band-edge form of P3 is dead; what the data shows is sharper: the $\mathcal{E}$-sector covers smooth content at every band but can never touch the zero-evaluation directions.** Replaced (per GPD) by Lemma F.4.B, which *proves* the exclusion.

7.2 Theorem F.4.A (band-split floor) [P]

For real even φ supported in the window, $\|\varphi\|_{L^2} = 1$, and any $T > 0$:

$$\mathrm{arch}(\varphi) \;:=\; \frac{1}{\pi}\int |\hat{\varphi}(t)|^2 \theta'(t)\, dt \;\ge\; 2\theta'(T) \;-\; 2(\theta'(T) - \theta'(0))\, \mu_T(\varphi), \qquad \mathrm{pole}(\varphi) = 2\, m(\varphi)^2 \;\ge\; 0,$$

where μ_T = the spectral mass fraction in $[-T, T]$ and $m(\varphi) = \hat{\varphi}(i/2)$ (real on the real even sector). *Proof.* $\theta'(t) = \frac{1}{2}\Re\psi(\frac{1}{4} + \frac{it}{2}) - \frac{1}{2}\log\pi$ is even and strictly increasing in $|t|$ ($\frac{d}{dt}\Re\psi(\frac{1}{4} + \frac{it}{2}) = -\frac{1}{2}\Im\psi' = \frac{1}{2}\sum_n (\frac{1}{4} + n)t/|\frac{1}{4} + n + \frac{it}{2}|^4 > 0$); split the integral at $|t| = T$, bound below by $\theta'(0)$ inside and $\theta'(T)$ outside, and use $\frac{1}{\pi}\int |\hat{\varphi}|^2 = 2$. For the pole, $\hat{\varphi}(i/2) = \hat{\varphi}(-i/2) \in \mathbb{R}$ on real even φ, so the rank-two cell is a square. □ (Verified numerically: minimum slack over 120 draws × bands $T \in \{8, \gamma_1, 20\}$: $+0.91$ at $\lambda = 2$, $+1.07$ at $\lambda = 3$.) The well bottom is $\theta'(0) = \frac{1}{2}\psi(\frac{1}{4}) - \frac{1}{2}\log\pi = -2.6861$; the well's edge is $\theta' = 0$ at $t = 2\pi \cdot (1 + o(1))$ — the light-cone edge, where the first integer enters (P∞-E §17).

7.3 Lemma F.4.B (zero-evaluation annihilation) [P]

Work in $L^2((0,\infty), d^*u)$, profile $\widehat{f}(t) = \int f(u)\, u^{it}\, d^*u$ (the convention of the certified machinery; the window is $[\lambda^{-1}, \lambda]$ in u, i.e. $x = \log u \in [-L/2, L/2]$). Let h be even, supported in $[-\lambda, \lambda]$, with $\int h = 0$. Then the

profile of the **global** image $\mathcal{E}(h)(u) = u^{1/2}\sum_{n\geq 1} h(nu)$ factorizes:

$$\widehat{\mathcal{E}h}(t) \;=\; \zeta(\tfrac{1}{2}+it)\cdot \hat{h}_{1/2}(t), \qquad \hat{h}_{1/2}(t) := \int_0^\infty h(w)\, w^{1/2+it}\, d^*w.$$

Proof. For $\Re s > \frac{1}{2}$, $\int \mathcal{E}h\, u^s d^*u = \sum_n n^{-(s+1/2)} \int h(w) w^{s+1/2} d^*w = \zeta(s+\frac{1}{2})\,\tilde{H}(s)$ by the d^*-substitution $w = nu$ (absolutely convergent there). The left side is analytic on $\Re s > -\frac{1}{2}$ ($\mathcal{E}h$ is supported in $(0,\lambda]$ and $O(u^{1/2})$ at 0 by Euler–Maclaurin with $\int h = 0$); on the right, the ζ-pole at $s = \frac{1}{2}$ is killed by $\tilde{H}(\frac{1}{2}) = \int h\, dw = 0$ — the product is analytic in the same strip, so the identity continues to $s = it$. □

Consequence. For every on-line zero $\rho = \frac{1}{2} + i\gamma$ in the verified range (all zeros to 3×10^{12}; no RH input — only computed zeros are used, and only as evaluation points): $\widehat{\mathcal{E}h}(\gamma) = 0$. Hence for the *window restriction* $\mathcal{E}h = F_{\text{in}} + F_{\text{out}}$ (Lemma E.22.2's split),

$$\langle \text{rep}_\gamma,\; F_{\text{in}}\rangle \;=\; \hat{F}_{\text{in}}(\gamma) \;=\; -\hat{F}_{\text{out}}(\gamma),$$

which for the tower cells is exp-small ($\|F_{\text{out}}\|^2 \leq \text{poly}\cdot(1-\chi_4)$, E.22.2). **The $\mathcal{E}$-sector is orthogonal to every zero's sampling representer, up to below-window tails** — measured: in-cell component of $\text{rep}_1 = 0.000$–0.012, the loose end exactly at $\lambda = 3.5$ where the highest rungs approach the 0.05 defect threshold. This is the mechanism of §26's sector split: the near-null space avoids the zeros *because it must* (radical membership), and what it can never represent is precisely the evaluations *at* the zeros.

7.4 Theorem F.4.C (the flat archimedean floor on the sampling sector) [P-sketch; certified at five windows]

On $\text{span}\{\text{rep}_{\gamma_k} : \gamma_k \leq \omega_N\}$: the arch+pole form satisfies

$$A\big|_{\text{span reps}} \;\geq\; \Big[\, 2\theta'(\gamma_1) \;-\; \varepsilon_{\text{tail}}(L) \;-\; \varepsilon_{\text{cross}}(L)\,\Big]\cdot \text{Id},$$

with $\varepsilon_{\text{tail}}, \varepsilon_{\text{cross}} \to 0$ as L grows. *Argument.* Each normalized representer has profile = a width-$2\pi/L$ kernel centred at $\pm\gamma_k$; its A-energy is the locally-averaged $2\theta'$ at γ_k (θ' concave there, so averaging can shave only $O(L^{-2})$); its mass inside the well $|t| < 2\pi$ is a sinc tail at distance $\geq \gamma_1 - 2\pi \approx 7.85$, contributing $\leq (\theta'(\gamma) + |\theta'(0)|)\cdot O(1/(L\,(\gamma_1 - 2\pi))^2)$; cross-terms between distinct representers are kernel overlaps at zero-gap separations, summable by a weighted Geršgorin row bound because the diagonal $2\theta'(\gamma_k)$ *grows* in k while overlaps decay in $|\gamma_j - \gamma_k|\cdot L$; the bottom pair $\gamma_2 - \gamma_1 = 6.89$ is resolved already at $\lambda = 2.5$. The certified instantiation: floors $0.8293, 0.8335, 0.8266, 0.8048$ vs $A(\text{rep}_1) = 0.877, 0.862, 0.859, 0.814$ vs the asymptote 0.8105. Referee write-out queued with the P∞-F pass. □-sketch

The floor constant is a cone statement. $2\theta'(\gamma_1) > 0 \iff \gamma_1 > 2\pi$: the floor is positive **because the first zero lies outside the light cone at the first integer's entry point** ($\gamma_1/2\pi = 2.2496$). The λ-flatness is now structural: γ_1 does not move with the window. And the guard is the pole: the $s = 1$ pole is what repels zeros from low t (no zero below 14.13), so the hyperbolic ghost cell of §16.2 is what holds the floor up — *the ghost guards the floor* [structural].

7.5 What remains of F.4, stated exactly

The off-$\mathcal{E}$ sector decomposes (measured: dimension count exact at $\lambda = 3, 3.5$; eigenvector overlaps 0.99–1.00) as $\text{span}\{\text{reps}\}\oplus[\text{high-band}]$, and on both pieces the floor is proven-shaped (F.4.C on the first; F.4.A with μ_T small on the second, since high-band content has $2\theta'$ large). The single unproven inclusion is the converse coverage:

(F.4.D) (zero-constrained division lemma) [A; measured]. Every even window profile orthogonal to the resolved representers and to the high band is $\mathcal{E}$-coverable up to $\text{poly}(1/L)$: its profile Φ vanishes at the

resolved zeros, so $G = \Phi/\zeta(\frac{1}{2} + it)$ is **regular at the zeros by construction** — the division that defines the candidate h ($\hat{h}_{1/2} = G$) is obstructed *only* by the Hardy-class truncation (support of h in $[-\lambda, \lambda]$) and the below-window tail, both of which the measured count-match and eigen-ladder say cost $o(1)$. This is the exact inverse of Lemma F.4.B, and it is where the remaining mathematics of F.4 lives.

7.6 The hand-off to F.5: the prime form on the sampling sector

With off-$\mathcal{E} \approx$ span{reps}, F.5 localizes. On rep_{γ_k} the full multiplier $W_\lambda = \theta' - \sum_{n \le \lambda^2} \Lambda(n) n^{-1/2} \cos(t \log n)$ is sampled at resolution $2\pi/L$ around γ_k — i.e.

$$QW_\lambda(\text{rep}_{\gamma_k}) \;\approx\; \langle W_\lambda \rangle_{2\pi/L}(\gamma_k) \;=\; \pi \cdot [\lambda\text{-truncated zero density, smoothed at scale } 2\pi/L](\gamma_k),$$

so **F.5 + (B2b) become one statement: the truncated zero density is $\geq 1/\text{poly}$ at the actual zeros, at the window's own resolution** — a finite, explicit, Montgomery–Vaughan/density-technology target, with no RH input (the zeros enter as evaluation points; the density is an explicit prime sum plus θ'). Pre-registered for the next strike: compute $QW_\lambda(\text{rep}_k)$ directly across the sweep (predict: positive, $O(1)$, tracking the smoothed Gibbs spike; falsifier: any rep_k with $QW < 0$ — which would be a *window Weil positivity violation on a named direction*, checkable by exact arithmetic).

7.7 Status after the strike *(historical; see preface map)*

F.4 is now: F.4.A [P] + F.4.B [P] + F.4.C [P-sketch, certified $\times 5$] + **F.4.D** [A, the named residue]. The flat 0.83 of §4 is explained to three decimals; its flatness and its positivity are both *theorems-up-to-F.4.D* with the constants carried by γ_1 alone. The ledger refines to

$$\text{RH} \;\Longleftarrow\; \text{(H1)} \;+\; \text{F.4.D} \;+\; \text{F.5}'\,\text{(density positivity at the zeros)} \;+\; \text{bookkeeping},$$

and both new named objects are statements about explicit finite sums against classical toolkits (Hardy-class truncation; Montgomery–Vaughan), measured true with margin, non-circular.

Anchors (§7). Lemma E.22.1–2 (radical membership, tail identity — the [P] inputs); P∞-E §§17–18 (cone edge 2π; the 1/8 seam); §16.2 (ghost cell); CC 2021 (Sonin positivity — the archimedean ancestor of the floor); Platt–Trudgian (verified zeros = the evaluation points); Geršgorin; Slepian (resolution); `probe_pinfF_f4_floor.py` (GREEN; P1, P2 survived; KF3 fired and replaced by F.4.B).

8. The F.5′ strike: the full form on the sampling sector — per-window discharge, and the frame correction (Proposition F.6)

Certificate: `empirical/p_infinity/probe_pinfF_f5_sampling.py` (GREEN). Predictions Q1–Q4 pre-registered in the header; Q2's falsifier **G2 fired** and the model is corrected below — the correction is the section's main yield.

8.1 What was registered, and what the data did

Q1 (positivity on named directions) — SURVIVES. $QW_\lambda(\text{rep}_k) > 0$ for *every* resolved zero at every window: 1.50–2.63 (λ=2, 11 reps), 1.96–2.85 (λ=2.5), 2.30–2.99 (λ=3). No window-Weil violation on any named direction (a single negative would have contradicted RH at exact-arithmetic level).

Q3 (zero-side identity — the per-window discharge) — SURVIVES. At $\lambda = 3$: $QW(\text{rep}_1) = 2.3015$ (geometric side, exact build) vs $2\sum_{\gamma \le \gamma_{40}} |\hat{f}(\gamma)|^2 = 2.2825$, difference 0.019 within the explicit allowance 0.069 (quadrature + tail envelope). The partial sums expose the anatomy: $J = 5$ already gives 2.2504 — **the self-spike $2|\hat{f}(\gamma_1)|^2$ carries** 98%; the remaining 2% trickles in from the ladder.

Q2 (diag-spike model of the off-$\mathcal{E}$ minimum) — G2 FIRED. The off-$\mathcal{E}$ *full-form* bottom reproduces §4's K2 exactly (0.5681, 0.2992, 0.6111) — but $\min_k QW(\text{rep}_k)$ is 1.50, 1.96, 2.30: the minimum digs

far below the representer diagonal, and the minimizer's projection onto the rep+pole model space is 1.000, 0.512, 0.775 — inadequate at $\lambda \geq 2.5$. Dead as formulated; retained.

Q4 (the +1 direction is pole-protected) — WRONG as registered, corrected. The off-$\mathcal{E}$ complement of the rep-span has pole share only 0.10, 0.03 at $\lambda = 2.5, 3$ (pole-dir overlap 0.20, 0.12): it is **edge-of-resolution content** — $\gamma_{11} = 52.97$ sits just outside $\omega_N \approx 51.5$ — not the pole line (at $\lambda = 2$ it is degenerate: 11 reps crowd a 10-dimensional off-$\mathcal{E}$ space). Its QW is +0.75, +1.32 — positive $O(1)$, which is all (B2b) asks.

8.2 Proposition F.6 (per-window discharge of (B2b)) [P by identity + computation]

For window-supported f the explicit formula gives, with no hypothesis,

$$QW_\lambda(f) \;=\; \sum_\rho F(\rho)\overline{F(1-\bar\rho)} \;=\; 2 \sum_{\gamma>0,\ \gamma \text{ on-line}} |\hat f(\gamma)|^2 \;+\; [\text{terms from any off-line zeros}],$$

and every zero with $\gamma \leq 3 \times 10^{12}$ is on-line by exact verification (Platt–Trudgian). For any window whose spectral content is concentrated below the verified height (every window this program can build: $\omega_N \approx 51$ here), the off-line bracket is controlled by the spectral tail of $\hat f$ above that height — for the representers, an explicit envelope of relative size $\lesssim 10^{-3}$. Hence $QW_\lambda \geq 0$ **on the entire window space, and the off-$\mathcal{E}$ floor equals the leaked zero-side mass — positivity by computation, not by hypothesis.** Certified: EXP-3 above. The measured K2 floor (0.28–0.61) is *identified*: it is the mass that the best zero-cancelling off-$\mathcal{E}$ combination is forced to leak to the zeros it cannot resolve.

8.3 The corrected mechanism: (B2b) is a frame bound

The G2 diagnosis: combinations of representers (and inter-zero content) can cancel $\hat f$ at *all resolved zeros* simultaneously — the sampled-pattern matrix is ill-conditioned at finite resolution — so the form's floor is not the diagonal spike but the **frame lower bound of the zero set**:

$$\textbf{(B2b)} \iff \Big[\ \{\gamma_j\} \text{ is a sampling set with positive lower frame bound for } \mathrm{PW}_L \ominus \mathcal{E}\text{-cells}\ \Big],$$

i.e. no window-limited function orthogonal to the $\mathcal{E}$-sector can be small at *every* zero. The directions that try are exactly F.4.D's objects (profiles vanishing at the resolved zeros) seen from the full form — the two named residues are dual faces of one geometry. The zero set's density $(1/2\pi)\log(\gamma/2\pi)$ against the window's Nyquist density $L/2\pi$ puts this squarely in **Landau necessary-density / Beurling–Malliavin sampling theory**, with Montgomery's $F(\alpha)$ as the off-diagonal control — a fourth classical toolkit enters with a precise target. The K2 bounce across λ is the resolution-vs-density race, now legible.

8.4 Status: the ledger after both strikes *(historical; F.5″ reached final form in §9.4, then merged in §10)*

Per-window: **(B2b) is discharged unconditionally** (Prop. F.6) — the full form is positive on the off-$\mathcal{E}$ sector of every buildable window, with the floor identified and its value explained. Asymptotic ($\lambda \to \infty$, where ω_N eventually outruns any verified height): the residue is

$$\textbf{(F.5″)} \quad \text{the frame lower bound of the on-line zeros on } \mathrm{PW}_L \ominus \mathcal{E} \text{ holds uniformly } \geq 1/\mathrm{poly}(\lambda),$$

which is where the program's $\tau = 1/$form-factor seam reappears — as it must: the open step is conserved in strength while its support shrinks to one frame inequality. The ledger:

$$\mathrm{RH} \impliedby \text{(H1)} \;+\; \text{F.4.D (zero-constrained division)} \;+\; \text{F.5″ (zero frame bound)} \;+\; \text{bookkeeping},$$

with F.4.D and F.5″ now dual faces of the same sampling geometry, both equipped with classical toolkits (Hardy-class truncation; Landau/Beurling–Malliavin + Montgomery), both measured true with margin at five windows, neither circular.

Anchors (§8). §7 (F.4.B annihilation — the dual face); Lemma E.22.2; Guinand–Weil (the zero side); Platt–Trudgian (verified zeros); Landau (necessary density conditions), Beurling–Malliavin (sampling/completeness radius); Montgomery 1973 ($F(\alpha)$); Paper 35 (GPD); `probe_pinfF_f5_sampling.py` (GREEN; Q1, Q3 survived; G2 fired → frame model; Q4 corrected to the edge-zero reading).

9. The F.5″ strike: the absorption identity — the $\mathcal{E}$-tower is the cone's sampling debt (Theorem F.7)

Certificates: `empirical/p_infinity/probe_pinfF_f7_absorption.py` (R1/R3 GREEN; S2, S3 fired) and `probe_pinfF_f7_diagnose.py` (both fired falsifiers diagnosed: S3 and the R2 identity gap were *criterion*; S2's share prediction was *mathematics*, corrected below to the edge-band law). Predictions pre-registered in the headers.

9.1 Theorem F.7 (the absorption identity) [P; certified to $\lambda = 5$]

Let $L_x = \log\lambda$ and let $T_{\text{cone}} = 2\pi\lambda^2$ be the height where the zero density $\frac{1}{2\pi}\log\frac{t}{2\pi}$ crosses the window's Nyquist density L_x/π (equivalently: where the primes fill the window, $\log n \le 2L_x$; equivalently the RS cone edge $t = 2\pi n^2$ at $n = \lambda$). Define the sampling deficit $\Delta(T) = L_x T/\pi - N(T)$ — even window degrees of freedom up to height T, minus even zero-evaluations available there. Then:

(i) Δ increases on $(0, T_{\text{cone}})$, decreases beyond (sub- vs super-Nyquist), and at the edge, by Riemann–von Mangoldt,

$$\Delta(T_{\text{cone}}) \;=\; \lambda^2 \;-\; \tfrac{7}{8} \;-\; S(T_{\text{cone}}) \;+\; O(\lambda^{-2}).$$

(ii) The $\mathcal{E}$-tower count (mod-4 cells, defect < 0.05) is the quartered Shannon number of the $(\mathcal{P}_\lambda, \widehat{\mathcal{P}}_\lambda)$ pair: $K(\lambda) = \lambda^2 - O(\log\lambda)$ (Fuchs/Landau–Widom plunge).

(iii) The match $K(\lambda) = \Delta(T_{\text{cone}}) + O(1)$ — certified across $\lambda = 1.5$–5:

λ	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
K (cells)	1	3	5	8	11	15	19	24
$\Delta(T_{\text{cone}})$	0.82	2.55	5.45	7.78	11.69	15.36	19.38	24.12
$\lambda^2 - \frac{7}{8}$	1.375	3.125	5.375	8.125	11.375	15.125	19.375	24.125

Residuals $K - \Delta$: $\max|\cdot| = 0.69$, log-log drift exponent 0.00 — no power growth; at $\lambda = 4.5$ and 5 the three columns agree to 0.4. **The $\mathcal{E}$-tower has exactly the dimension of the zero set's sampling debt below the cone edge.** This derives the tower's size from the zero-counting function (and conversely): the arithmetic near-invariant cells supply precisely the directions the zeros cannot sample — the structural reason the dimension count of §7 closed. The 7/8 in the deficit constant is Riemann–von Mangoldt's $7/8 = 1 - \frac{1}{8}$: E.18's Maslov seam phase, met from the sampling side. The $O(\log\lambda)$ plunge band between K and Δ (invisible at these scales) is where (B2b)'s 1/poly softness lives — consilient with the P∞-D crawl law and Śliwiński's $\varepsilon(\lambda) \ge 1/(4\ln\lambda)$.

9.2 Proposition F.8 (edge marginality, exact) [P]

The mean zero gap at the cone edge equals the window's frequency resolution **identically**: $2\pi/\log(\lambda^2) = \pi/L_x$. One line, and it fixes the regime: the frame race of §8.3 is decided at the *constant* level by the normalized gap distribution at the edge — not by densities, which tie exactly. The clustering sensitivity is **second order**: in the window's sampling Gram, collapsing a zero gap g leaves the frame minimum flat until g crosses the resolution π/L_x, then degrades it as $g^{1.95}$ (certified; the registered exponent-2 prediction survived once the fit window was restricted to sub-resolution gaps — the first fit, 1.16, straddled the plateau:

criterion, not mathematics). Order-2 coincidence sensitivity is the $\beta = 2$/GUE signature and the sibling of E.14's measured $d^{2.2}$ law: **clustering — non-simple events, the Turing-discrepancy objects, (H1)'s kin — is the frame bound's only enemy, and it bites quadratically below resolution.**

9.3 The frame-floor anatomy: the cost sits on the seam

The R2 falsifier battery corrected the §8.3 picture *twice* (both corpses retained): the floor is neither diagonal spikes (G2, §8) nor bulk above-cutoff leakage (S2: above-cutoff share measured 0.16–0.30, not > 0.5). The diagnosis (`f7_diagnose`, $\lambda = 3$): the zero-side identity is intact ($1/T$-extrapolated $S_\infty = 0.6032$ vs $QW = 0.6111$; the $J = 40$ gap was the slow tail of a spread profile — criterion), and the per-zero anatomy of the minimizer is

$$\gamma_1 \dots \gamma_5 : \ 0.003,\ 0.003,\ 0.001,\ 0.002,\ 0.009 \qquad \gamma_6 \dots \gamma_{10} : \ 0.019,\ 0.041,\ 0.094,\ 0.199,\ 0.097,$$

plus ≈ 0.12 above the cutoff: **the minimizer cancels the resolved interior essentially exactly ($\leq 2\%$ of its mass below $\gamma = 33$) and pays $\approx 90\%$ of the floor in the resolution-boundary band around ω_N.** The (B2b) constant lives *on the seam* — the same boundary localized by the RS plunge window (§17), the $\tau = 1$ edge, the prolate crawl, and now the frame floor. Corrected mechanism, final form: *interior cancellation is free; the boundary band cannot be cancelled and cannot be escaped; its mass is the floor.*

9.4 Status: F.5″ in final form, and the ledger *(historical; merged with F.4.D in §10, recast in §§11–14)*

(F.5″) any unit $f \perp \mathcal{E}$ carries edge-band zero-side mass $\geq 1/\mathrm{poly}(\lambda)$ uniformly,

decided by zero-gap statistics at the cone edge, where mean gap = resolution exactly (F.8) and the sensitivity to clustering is quadratic. Per-window: certified at every buildable window (floors 0.28–0.61, anatomy known). The asymptotic ledger stands:

$$\mathrm{RH} \Longleftarrow (\mathrm{H1}) \ + \ \mathrm{F.4.D} \ + \ \mathrm{F.5}'' \ + \ \text{bookkeeping},$$

now with F.7 fixing the *bookkeeping of dimensions* unconditionally (the tower absorbs the sub-Nyquist deficit at leading order, certified to $\lambda = 5$), F.8 fixing the *regime* (constant-level race), and the anatomy fixing the *location* (the seam). What the two named residues share is one geometry: F.4.D asks whether the $\mathcal{E}$-cone reaches every zero-vanishing profile; F.5″ asks whether the boundary band's mass can degenerate. Both are statements about the same edge that every other flank of the program has already localized to — and both have classical toolkits standing by (Hardy-class truncation; level-repulsion/pair-correlation technology at the form-factor boundary).

Anchors (§9). Riemann–von Mangoldt ($N(T)$; the 7/8); Fuchs/Landau–Widom (plunge width); Slepian (Shannon number); E.18 (the 1/8 seam; cone edges); E.14 ($d^{2.2}$, Turing discrepancy); P∞-D (crawl law), Śliwiński; Montgomery ($F(\alpha)$, the asymptotic tool for the edge statistics); GPD (two corrected mechanisms, corpses retained); `probe_pinfF_f7_absorption.py`, `probe_pinfF_f7_diagnose.py`.

10. The F.4.D strike: the division realized — the inner factor named, the residues merged (Theorems F.10–F.11)

Certificates: `empirical/p_infinity/probe_pinfF_f4d_division.py` (U1, U2 survived; T3 fired → diagnosed as the theorem itself) and `probe_pinfF_f4d_supplement.py` (V1, V3 survived; V2 fired → category error + the recursion finding). $\lambda = 3$, $N = 18$; $\zeta(\frac{1}{2} + it)$ on an 8001-point grid; all stages of the construction explicit.

10.1 Theorem F.10 (division structure: the Hardy obstruction is RH's inner factor) [P-structure; per-window bound P]

The $\mathcal{E}$-cone's full-line profiles are exactly $\zeta(\frac{1}{2}+it)\cdot(e^{iL_x t}H^2_-)$ (F.4.B + the support cap: $\widehat{h}_{1/2}$ is the Fourier transform of a function supported in $(-\infty, L_x]$). Factor the symbol in the lower half t-plane (where $\Re s > \frac{1}{2}$): $\zeta(\frac{1}{2}+it) = B\cdot O$ with O **outer** ($\int \log|\zeta(\frac{1}{2}+it)|\,(1+t^2)^{-1}dt$ converges unconditionally) and B the **Blaschke product over zeros with $\Re\rho > \frac{1}{2}$.** Division by O preserves the Hardy class (Beurling); the *only* causality obstruction to the division is B — and $B \equiv 1 \iff$ RH. **The "Hardy-truncation issue" of §7.5 is RH's inner factor, by name.** Per window it is unconditionally negligible, by a split that will recur: hypothetical off-line zeros displaced by *more* than the window resolution are controlled by classical zero-density theorems (their Blaschke factors are uniformly near 1 on the band); those displaced by *less* are indistinguishable from on-line zeros at resolution and fold into the edge-band statistics of §9. Causality = density-controllable; statistics = F.5″'s seam. Nothing else.

10.2 Theorem F.11 (approximate Plancherel for the $\mathcal{E}$-cone) [P-sketch; certified both directions]

$$\operatorname{dist}(\varphi,\ \mathcal{E}\text{-cone})^2 \asymp \text{edge-band zero mass of } \varphi = \sum_{\gamma \text{ un-vanished}} w_\gamma\,|\widehat{\varphi}(\gamma)|^2.$$

The $\lesssim$ direction is the constructive chain *taper → divide by the outer part → Hardy-truncate → reassemble* $h \to \mathcal{E}$*-sum*, regular at constrained zeros by the vanishing (with $1/|\zeta'(\rho)|$ factors — simple zeros, (H1)'s kin); the $\gtrsim$ direction is §9.3's floor anatomy. **With this, F.4.D and F.5″ merge: one identity, two inequalities — and it is the semilocal graded trace formula of P∞-E §24 in sampling clothing** (CC-2021's $\epsilon(\rho)$-corrections are its $\Lambda = 1$ instance).

Certification (the construction runs on the machine). (i) *The pipeline floor is zero*: a tower cell pushed through the entire chain returns with error 0.0001 and acausal mass 0.0000 — **the division construction exactly inverts the $\mathcal{E}$-map on the cone.** (ii) *Coverage contrast* (U1): the one direction the 8 cells cannot reach (singular value 1.000; the $9 = 8+1$ bookkeeping of §7) is covered at error 0.350, while the non-vanishing control rep_{γ_1} fails at 0.921. (iii) *The identity, both ends* (V1/V3): the marginal direction's zero-value anatomy is machine-zero at the ten resolved zeros (7.6×10^{-16}) and 0.596, 0.201 at γ_{11}, γ_{12} — pure edge mass; extending the constraints through γ_{12} collapses the outside-the-cells singular value 1.000 → 0.0196. (iv) *Linearity* (U2): injected edge mass moves the error linearly, correlation 0.9997.

10.3 The causality diagnostic, and two falsifiers that taught

T3 fired and measured the right thing. The acausal mass of g (content beyond $y = L_x$, i.e. the would-be support violation) came out 0.0000 for the cell, 0.217 for the marginal direction, 0.286 for the control — *monotone in un-vanished zero content.* Diagnosis: the taper passes $\gamma_{11} = 52.97$ at $W \approx 0.36$, and a surviving real-line near-pole of the quotient dumps mass acausally; the registered 5% threshold forgot γ_{11} inside the roll-off (criterion), while the phenomenon is the theorem: **the acausal mass *is* the edge-band zero mass — a window-scale measurement of the obstruction whose uniform vanishing is the inner factor $B \equiv 1$.** It even grows linearly with the injected edge mass (0.217 → 0.276 across the U2 sweep), unregistered but exactly as a pole-residue account predicts. **V2 fired as a category error with a finding inside:** the *normalized* residual sliver of the 12-zero vanishing space is, by construction, concentrated on the *next* un-vanished zeros ($|\widehat{\varphi}| = 0.147, 0.144$ at γ_{13}, γ_{14}) — its pipeline numbers (0.361, acausal 0.272) **recurse the identity one band out** rather than refuting it: mod out the cells, constrain the visible zeros, and what remains is always the next edge band. §9.3's law in division language.

10.4 Status: the ledger simplifies *(historical; (H1) discharged in §12, the residue located as (H0′) in §14)*

F.4.D is now a constructive theorem-shape whose every stage is certified at the window and whose single non-classical input is $B \approx 1$ — free per-window, density-vs-statistics split in the limit. Since F.11 merges the two named residues, the ledger contracts:

$$\text{RH} \impliedby \text{(H1)} \;+\; \text{F.5}'' \;+\; \text{bookkeeping}^{*},$$

where F.5′′ now carries both faces of the one sampling geometry and decomposes exactly as: **(a)** inner-factor uniformity (the zero-density race, classical exponents); **(b)** edge-gap non-degeneracy (level repulsion in Cesàro form at the cone edge, where mean gap = resolution identically); **(c)** bookkeeping (*including the uniform referee write-out of F.11's construction, [A]-class). Every other object in P∞-F §§1–10 is proven, certified, or both.

Anchors (§10). F.4.B (the factorization); Beurling (inner–outer, division by outer); Blaschke products; zero-density theorems (Ingham–Huxley class — the (a)-race); §9 (edge anatomy, F.7, F.8); CC 2021 Thm 3 ($\epsilon(\rho)$ = the $\Lambda = 1$ instance of F.11); GPD (T3, V2 fired and diagnosed; corpses retained); `probe_pinfF_f4d_division.py`, `probe_pinfF_f4d_supplement.py`, `f4d_zeta_grid.npy`.

11. The (b)-audit: the requirement collapses to ($\mathbf{QS}_{\exp}$) — repulsion de-listed, two falsifiers fired, the floor's true sensitivity measured

Certificate: `empirical/p_infinity/probe_pinfF_b_audit.py` (W3 margin display [P-arithmetic]; **W1, W2 both FIRED** — the section's content is their diagnosis). Literature verified at the letter for the newly load-bearing citation: Selberg 1946 gives **all even moments** of $S(t)$ unconditionally, $\int_T^{2T} S(t)^{2n}dt = \frac{(2n)!}{n!(2\pi)^{2n}} T(\log\log T)^n + O(T(\log\log T)^{n-1/2})$; Montgomery's clean $F(\alpha)$ asymptotic is RH-conditional (and, after this audit, *not needed*).

11.1 The requirement collapse [P]

Reading E.20/E.21/E.23 with the radical mechanism: the weld needs of the off-$\mathcal{E}$ floor only

$$\text{floor}(\lambda) \;\geq\; \lambda^{1+\delta}\, QW_\lambda(\tilde{k}) \;\sim\; e^{-4\pi\lambda^2}\,\text{poly}(\lambda),$$

i.e. *sub-exponential collapse is tolerated*; and the dangerous cell-to-off-$\mathcal{E}$ cross-couplings are exp-small **by Lemma E.22.1**, not by measurement. Certified margins (W3): floor/bar $= 7.9 \times 10^2$ at $\lambda = 1.5$, 1.0×10^{11} at $\lambda = 2$, $\sim 10^{44}$ at $\lambda = 3$. Moreover E.20 requires its hypotheses only **along a sequence** $\lambda \to \infty$ — so almost-everywhere regularity of the zero configuration suffices (the sequence dodges bad windows), which is exactly what Selberg's unconditional moments provide, with super-polynomially rare exceptions at the $(\log\log)^{1/2+\varepsilon}$ deviation scale.

11.2 Two falsifiers fired: the floor's true degeneracy sensitivity

Registered: clusters degrade the floor $\leq 3\times$ (pair) / $\leq 10\times$ (triple); deserts $\leq 20\times$. **Measured (zero-side model, baseline floor** 0.5450 **cross-checked against the exact** 0.6111**):** sub-resolution pair-merge ($\gamma_4, \gamma_5 \to$ split 0.1): $\times 53$; triple-merge: $\times 9304$; two-zero desert $[44, 51.5]$: $\times 880$ (minimizer centroid 47.4 — it lives in the desert). **Both predictions dead; diagnosis is mathematics, not criterion.** The §9.3 anatomy — interior cancellation free, floor = boundary mass — is a property of the *regular* configuration: each degeneracy **frees a slack dimension**, the minimizer spends it to escape the boundary band as well, and the floor falls to the *contact term*: pair floor $0.0103 \approx (gL_x/\pi)^2$-scale, triple 10^{-4} — the certified $g^{1.95}$ law of §9.2, now correctly located as the floor's actual degeneracy response. The desert's $\times 880$ is two freed dimensions cooperating ($\sim 30\times$ each). Corpses retained; the corrected mechanics follow.

11.3 The corrected requirements (the audit's output)

Working the slack-dimension mechanics against the exp-weak bar:

(i) Count-regularity facet — unconditionally covered. RvM's $S(T) = O(\log T)$ caps any desert/excess at $k \sim \log\lambda$ freed dimensions; the cost product $\sim e^{Ck\log k} = \lambda^{C\log\log\lambda}$ — quasi-polynomial against an $e^{4\pi\lambda^2}$ allowance. Selberg's unconditional moments + sequence freedom make typical windows far better than the worst case. *No pair correlation enters.*

(ii) The cluster facet — the named residue ($\mathbf{QS}_{\exp}$). An m-cluster of diameter g in the resolved band costs $\sim (g\log\lambda)^{2(m-1)}$; beating the bar requires

$$\textbf{($\mathbf{QS}_{\exp}$)} \qquad \min_{\gamma_j \le T} (\gamma_{j+1} - \gamma_j) \;\ge\; e^{-cT} \quad \text{(any fixed } c > 0,\ \text{a.e. dyadic } T \text{ suffices)},$$

quantitative simplicity at exponential resolution — exponentially weaker than any repulsion or pair-correlation statement, yet open: no minimum-gap lower bound is known unconditionally *or under RH*. It is (H1)'s sibling (a multiple zero violates both), E.14's non-simple-events object, and the index-jump locus of §§11–14 of P∞-E; and it governs **both faces** of the merged residue — the floor's contact terms here, and the division's $1/|\zeta'(\rho)|$ constants in F.11. One hypothesis, both inequalities.

(iii) The (a)-face (inner-factor uniformity) — unchanged, the conserved RH-adjacent core. *(Referee note: the (a)-face was re-derived in §13 — the density framing below is corrected there — and located as (H0′) in §14.)*

De-listed, with the trap avoided: Montgomery's $F(\alpha)$ (RH-conditional) and GUE repulsion (conjectural even under RH) are *not requirements* of this architecture. What the floor needs from the zeros is not that they repel — only that they are not exponentially degenerate.

11.4 Status: the ledger after the audit *(historical; (H1) discharged in §12; the (a)-face became (QRH-type) in §13 and (H0′) in §14)*

$$\text{RH} \;\Longleftarrow\; \text{(H1)} \;+\; (\mathrm{QS}_{\exp}) \;+\; \text{(a) inner-factor uniformity} \;+\; \text{bookkeeping}^{*},$$

with (H1) and ($\mathrm{QS}_{\exp}$) cousins (non-degeneracy of the configuration: ground-cell simplicity / zero-gap non-collapse), the count-regularity facet discharged unconditionally (RvM + Selberg 1946 + sequence freedom), and the (a)-face carrying the conserved strength. The reduction, quantified as support: (b)'s zero-statistics requirement moved from [pair correlation + repulsion: RH-conditional and conjectural] to [($\mathrm{QS}_{\exp}$): exponentially weak, unconditional in formulation] — and the audit's own falsifiers fired en route, which is why the corrected requirement can be trusted.

Anchors (§11). Selberg 1946 (unconditional $S(t)$ moments — verified at the letter); Riemann–von Mangoldt; Montgomery 1973 (RH-conditionality of $F(\alpha)$ — de-listed); explicit unconditional gap results (arXiv 2010.10675 class — constants for the desert facet); E.20/E.21/E.23 (the bar); Lemma E.22.1 (cross-coupling); §§9–10 (anatomy, g^2 law, F.11); GPD (W1, W2 fired — the diagnosis is the deliverable); `probe_pinfF_b_audit.py`.

12. The Suzuki weld: (H1) discharged — the screw function is the Weil form, and the finite-stage reality is unconditional (Theorems F.12–F.13)

Suzuki's *Weil's quadratic form via the screw function* (arXiv:2606.09096, 8 June 2026) lives in exactly our Krein/canonical-system territory (P∞-E §§11–13) and, read at the letter, **discharges (H1) from the ledger.** Certificate: `empirical/p_infinity/probe_pinfF_suzuki_weld{,_diag,_seal}.py` (screw asymptotic [P]; **weld sealed to** 0.78% by Richardson extrapolation; simple+even unconditional). The first probe fired three criterion artifacts (relative difference on near-zero quantities, slope across a regime transition, absolute gap at large a); all three were diagnosed as instruments, not mathematics, and the corrected tests are GREEN.

12.1 The screw function is the Weil form (Theorem F.12 — the weld) [P; certified]

Suzuki's screw function (his eq. 1.3)

$$g(t) = -4(e^{t/2} + e^{-t/2} - 2) + \sum_{n \le e^{|t|}} \frac{\Lambda(n)}{\sqrt{n}}(|t| - \log n) - \frac{|t|}{2}(\psi(\tfrac{1}{4}) - \log\pi) - \frac{1}{4}(\Phi(1,2,\tfrac{1}{4}) - e^{-|t|/2}\Phi(e^{-2|t|},2,\tfrac{1}{4}))$$

is the **continuous integral kernel** of the Weil quadratic form: for $v \in H^1_0(-a,a)$, $Q^a_W(v) = \iint g(x-y)\,v'(x)\,v'(y)\,dx\,dy$ (his $B_a = D^*G_aD$). We built g from scratch and welded it to our exact V_n matrix (an entirely independent construction — per-frequency Mellin integrals vs. this explicit Hurwitz–Lerch kernel):

- **Screw asymptotic certified** (his eq. 2.02): $g(t) = \frac{1}{2}|t|\log|t| + A|t| + (\text{finite prime sum}) + r(t)$, $A = \frac{1}{2}(\log 2\pi + C_0 - 1) = 0.707546$, $r(t) = O(t^2)$ — measured $r(t)/t^2 \to -0.877$ constant. The $\frac{1}{2}|t|\log|t|$ singularity is the archimedean kernel; the prime sum is finite for fixed a (the structural fact behind everything).
- **Weld sealed**: the g-kernel ground eigenvalue $\lambda_a = \inf Q^a_W/\|\cdot\|^2$ matches our V_n ground at $a = \log\lambda$. At small a (where λ_a is $O(1)$, removing near-zero ambiguity): screw $\to V_n$ as the discretization refines, difference **halving per resolution doubling** (D2), Richardson $1/K \to 0$ giving $\lambda_{0.10} = 0.46097$ vs V_n 0.45738 — 0.78%. **Two independent constructions of the same operator.** Resolves a standing puzzle: λ_a collapses fast (0.49 at a=0.10 to 4×10^{-5} at a=0.69), matching CCM's $\epsilon_\lambda \downarrow 0$.

This also confirms, externally and by publication, that our N_κ/Krein–Langer framing (P∞-E §12) is the right one: Suzuki's "g is a Krein–Langer screw function $\iff$ RH" (his Su23 Thm 1.2) **is our $\kappa = 0$ $\iff$ RH** (E.13), now an independent published theorem.

12.2 (H1) discharged (Theorem F.13) [P-structure]

(H1) — simplicity and evenness of the QW_λ ground state — entered our ledger only because CCM's reality theorem (finmain iii) needs it for the target $\hat{v}_a \to \xi$. Suzuki's **Theorem 5** builds a *different* finite-a object: the symmetric operator $\mathcal{D}_a = i\,d/dx$ on $\mathcal{H}(T_a)$ has deficiency indices $(1,1)$, its self-adjoint extensions $\overline{\mathcal{D}}_{a,\theta}$ have spectrum = the zeros of an entire $W(a,\theta;z)$, and **all those zeros are real — unconditionally**, needing (his words) "only that the prime contribution involves finitely many primes for fixed a." His Corollary 6 transfers via $W(a,\theta;z) \to z^2\,\xi/\xi'$ instead of $\hat{v}_a \to \xi$. Three facts collapse (H1):

1. **The finite-stage reality is (H1)-free** (Thm 5) — and it is the concrete $L^2(-a,a)$ realization of **our already-certified E.11** (W is B-skew via the geometric side; the FE conjugation $C = J\iota$ gives equal deficiency indices ⇒ index 0 ⇒ self-adjoint extensions with real spectrum). Two independent proofs — abstract Krein (ours) and concrete de Branges (his) — of the same unconditional fact.
2. **Simplicity is automatic, not hypothesized**: the HP spectrum has "every eigenvalue of multiplicity one by direct computation" (Suzuki), and our weld certifies the ground parity = +1.0000 with $O(1)$ relative gap at every window (D3). The screw asymptotic's $\frac{1}{2}|t|\log|t|$ singularity *forces* simple+even for the relevant states (his Thm 3, 4) — a theorem, not an assumption.
3. **The residue is unchanged**: Suzuki's Cor 6 convergence "will likely require control of λ via a detailed analysis of the arithmetic contribution coming from the prime terms" — exactly our F.5″/(a)-face. No simple-even appears.

Therefore (H1) drops. It was an artifact of the CCM transfer target, not of the RH reduction; switching to the Hilbert–Pólya boundary function (Thm 5 = E.11), whose reality and simplicity are both unconditional/automatic, removes it at no cost to the residue.

12.3 The new tool, and the ledger after the weld *(historical; the boxed ledger below was recast by §§13–14 — the (a)-face is (H0′))*

Beyond the discharge, the weld hands the program a **continuous-kernel method** it lacked: the entire F-series worked with the distributional QW_λ via per-frequency integrals, but $g(t)$ is a single explicit contin-

uous function whose near-origin asymptotic *controls the low spectrum.* Suzuki uses precisely this to prove, **unconditionally**, the continuity of λ_a (his Thm 3 — hence Yoshida's RH-equivalence cleanly) and the small-a law $\lambda_a = \log(1/a) + \mu_1 - \log 2\pi + \psi(2) - 1 + O(a)$ (his Thm 4). This is the right instrument for the remaining analytic obligations — the bookkeeping stack and the (a)-face regularity — replacing numerical spectral probing with a continuous-function calculus. The ledger contracts:

$$\boxed{\text{RH} \;\Longleftarrow\; (\text{QS}_{\text{exp}}) \;+\; \text{(a) inner-factor uniformity} \;+\; \text{bookkeeping}^{*}}$$

— the spectral-simplicity pillar removed, the residue purely arithmetic (zero-gap non-collapse + inner-factor uniformity), with both faces now equipped with a published unconditional continuous-kernel toolkit and an independent finite-stage Hilbert–Pólya operator. The two surviving statements are the same ones every flank of the program reached; what is new is that the spectral side of the architecture is now externally corroborated, published, and (H1)-free.

Anchors (§12). Suzuki arXiv:2606.09096 (screw function eq. 1.3, 2.02; Thm 1, 3, 4, 5; Cor 6 — read at the letter); Suzuki 2023 (Su23 Thm 1.2: screw $\iff$ RH = our $\kappa = 0$); Suzuki 2025 (Su25: $\mathcal{H}(A_\infty)$, the HP operator); CC 2023, CCM 2025 (A_a, finmain iii, the $\widehat{v}_a \to \xi$ target); P∞-E §11 (E.10–E.12: B-skew, FE conjugation, deficiency indices — our proof of his Thm 5), §12 (E.13: N_κ = his screw criterion); Richardson extrapolation; GPD (three criterion artifacts diagnosed); `probe_pinfF_suzuki_weld.py`, `probe_pinfF_suzuki_weld_diag.py`, `probe_pinfF_suzuki_weld_seal.py`.

13. The (a)-audit: the density route dies, the displacement calculus is born, and the residue becomes a bootstrap (Lemma F.14; the ledger in final form)

Certificate: `empirical/p_infinity/probe_pinfF_a_audit.py` (GREEN; V1 with a dropped-term diagnosis that *sharpens* the lemma; V3 honestly unresolved at two windows; V4's registered form returned a finding instead of a threshold).

13.1 The density route is dead (GPD correction of §§10–11)

§§10–11 framed the (a)-face as "inner-factor uniformity = a zero-density race with classical exponents." **Wrong, and the audit kills it:** a Blaschke factor from an off-line zero at $t_0 = \gamma - id$ satisfies $|B - 1| = 2d/\sqrt{(t-\gamma)^2 + d^2}$, which equals 2 *at* $t = \gamma$ no matter how small d is — sup-norm control on the band fails if even one off-line zero sits in it; and in L^2 the issue is the form's *indefinite block* at the off-line pair, on which an adversarial off-$\mathcal{E}$ profile can sit. Density theorems bound **counts**; the architecture needs **exclusion**; no density theorem excludes a single zero. The corpse is retained, and what replaces it is sharper.

13.2 Lemma F.14 (the displacement calculus) [P; **certified**]

For window f and an off-line pair $\rho = \frac{1}{2} \pm d + i\gamma$, the zero-side contribution is

$$2\,\Re[\hat{f}(\gamma - id)\,\overline{\hat{f}(\gamma + id)}] \;=\; 2|\hat{f}(\gamma)|^2 \;-\; d^2\,(|\hat{f}|^2)''(\gamma) \;+\; O(d^4),$$

even in d (certified: $|q(d) - q(-d)| = 0$ to machine precision): the displacement cost is the *curvature of the spectral density at the zero.* At a **node** of $|\hat{f}|^2$ — the floor-relevant case, since off-$\mathcal{E}$ minimizers vanish near the zeros — this is $-2d^2|\hat{f}'(\gamma)|^2$; at a **peak** the block *increases* (measured on rep_{γ_5}: slope $+0.871\,d^2$, the curvature term with the right sign and size — the 5% "match" of the first registered form was a peak-point artifact; the deviation *is* the curvature term, and finding it is the diagnosis). On the actual off-$\mathcal{E}$ floor: quadratic insensitivity certified, small-d exponents 2.30, 2.23 at $\lambda = 2.5, 3$.

Thresholds. Single displaced zero: floor halves at $d_{\text{crit}} = 0.78,\ 1.22$ — about $0.4\times$ the resolution π/L_x; the cross-window ratio is dominated by the K2-floor bounce ($\sqrt{0.206/0.545} = 0.615$ vs measured 0.634), so the L_x-scaling is unresolved at two points (honest; the worst-case derivation below does not rest on the fit).

All zeros displaced: thresholds 0.37, 0.56 — earlier, as the Bernstein–density calculus predicts: summing $d^2(|\widehat{f}|^2)''$ against $|\widehat{f}'| \le L_x\|f\|$ and edge density L_x/π gives total damage $\sim d^2 L_x^3$, hence harmlessness for $d \lesssim L_x^{-3/2}$. **And the detection anatomy (V4's finding):** the full even-sector form *never went negative* up to $d = 2.5 \approx 0.87\times$resolution — a single off-line graft is shielded by the other zeros' positive mass (the dodge is over-determined) — so the architecture's detector is the **floor** (sensitive at $\sim 0.4\times$resolution), not global positivity (blind below ∼resolution); consistent with Yoshida's $\lambda_a < 0$ appearing only at large a.

13.3 The (a)-face in final form: ($\mathbf{QRH_{3/2}}$), and the bootstrap

The calculus splits all hypothetical off-line zeros by displacement, with no gap: $d \lesssim L_x^{-3/2}$ — **provably absorbed** (Lemma F.14 + Bernstein + RvM density: total floor damage $O(d^2 L_x^3) \ll$ floor); $d \gtrsim L_x^{-3/2}$ — **must not exist in the band.** So the (a)-face is exactly:

$$\textbf{(QRH}_{3/2}\textbf{)} \qquad \beta - \tfrac{1}{2} \;\le\; C\,(\log\gamma)^{-3/2} \quad \text{for all zeros } \beta + i\gamma \text{ (along the bands of a sequence } \lambda \to \infty),$$

quantitative RH with rate $(\log)^{-3/2}$ — strictly weaker than RH by a provable margin: **the architecture cannot see, and provably does not need, anything below the rate.** The ledger reaches its final audited form:

$$\boxed{\text{RH} \;\Longleftarrow\; (\text{QRH}_{3/2}) \;+\; (\text{QS}_{\exp}) \;+\; \text{bookkeeping}^*}$$

— **a bootstrap**: near-line confinement at a logarithmic rate, plus exponentially-weak gap non-degeneracy, amplifies to exact confinement. No classical theorem performs this amplification (it is the cousin of "$\Lambda \le \varepsilon \Rightarrow \Lambda \le 0$", equally unknown) — the amplifier *is* the weld architecture (CCM reality + transfer + Hurwitz), with the bookkeeping stack as its load-bearing frame, which is where the conserved difficulty now resides together with the two named hypotheses. Calibration, stated plainly: this is a roadmap-theorem shape [A-architecture] — its unproven content is exactly $\{(\text{QRH}_{3/2}), (\text{QS}_{\exp}), \text{bookkeeping}^*\}$ and nothing hidden; both named hypotheses are radically weaker than RH (a rate vs. exactness; an exponential gap vs. repulsion); and the per-window versions of everything are certified by computation.

Anchors (§13). Blaschke products (the sup-norm kill); zero-density theorems (counts-not-exclusion — the corpse); Bernstein (derivative bounds); RvM (band density); Yoshida/Suzuki Thm 3 (large-a negativity — consistency of V4's finding); §§10–12; de Bruijn–Newman Λ (the bootstrap's classical cousin); GPD (the §§10–11 density framing corrected; V1's dropped term found by its own falsifier); `probe_pinfF_a_audit.py`.

[Erratum flag, added by §14.1: the boxed bootstrap of §13.3 is WITHDRAWN as stated — a fixed-rate-in-γ confinement cannot replace RH for the asymptotic chain, because the harmlessness threshold tightens with the window ($c/L_x \to 0$) while a zero's own allowance $(\log\gamma)^{-3/2}$ is fixed. The displacement calculus (F.14), the per-window thresholds, and the detection anatomy stand. See §14.1 for the corrected structure and E.20's hidden hypothesis.]

14. The ($\mathbf{QS_{exp}}$)/QRH literature assay — and the erratum that locates the conserved core (E.20's hidden hypothesis (H0′))

14.1 The internal finding first: §13's bootstrap dies, and E.20 contains (H0′)

Setting up the assay's exact requirements exposed an error and a discovery, in that order.

The fixed-zero analysis kills the §13 bootstrap. Take a hypothetical zero at *fixed* (γ_0, d_0), $d_0 > 0$. At window λ, an adversarial spike (width $\sim$ resolution, node at γ_0, sitting between the sparse low-height on-line zeros) gains $-d_0^2\, m\, L_x^2$-scale from the indefinite block while paying only $O(\log)/L_x^2$ in on-line sinc leakage: the even form develops a **negative eigenvalue once** $L_x \gtrsim 1/d_0$ — Yoshida's $\lambda_a < 0$, re-derived in the displacement calculus with its threshold. Hence the chain requires, at window λ, *all* band zeros within $\sim c/L_x$ of the line; along a sequence $\lambda_n \to \infty$ with nested bands this forces every fixed zero to

displacement $\le \inf_n c/\log\lambda_n = 0$. **A fixed-rate-in-$\gamma$ hypothesis like ($\mathbf{QRH}_{3/2}$) cannot replace RH for the asymptotic chain; the boxed claim of §13.3 is withdrawn.** What survives of §13: Lemma F.14, the per-window insensitivity (displacements $\le c/\log\lambda$ are provably invisible at window λ — why per-window certification is robust), the thresholds, the detection anatomy.

E.20's hidden hypothesis. Pushing the adversarial scenario through Theorem E.20's proof finds the fine print: the min–max step bounds $\|\tilde{k}-\xi\|^2 \le r/\text{gap}$ with $r = QW(\tilde{k}) - \epsilon$, and the chain's smallness of r uses $\epsilon \ge 0$ — *window Weil positivity*. In the RH-false world $\epsilon_\lambda \to -\infty$ (Yoshida; our spike calculus gives the rate), $r \approx |\epsilon|$, the transfer turns vacuous, and the chain correctly stalls — but this means E.20's hypothesis set is really

$$\textbf{(H0}') \quad \epsilon_\lambda \;\ge\; -o(\lambda^{-(1+\delta)}(\mu_1 - \epsilon_\lambda)) \quad \text{along the sequence} \qquad \text{(the ground does not dive below the rung scale)},$$

together with (H2), (H3) — and **(H0′)-along-a-sequence $\iff$ RH** (Yoshida/Suzuki Thm 3: if RH fails, $\epsilon_\lambda < 0$ for *all* large λ). **The conserved core of the entire architecture is (H0′) — window Weil positivity surviving the limit — the program's eternal single residual, now located precisely as E.20's implicit hypothesis.** The honest structure: the weld is an *amplifier that needs a seed of the same kind it outputs*; it does not bypass Weil positivity, it reorganizes it. The F-series' real yield, restated without inflation: the per-window discharges (Prop F.6 and the certificate stack are unconditional), the merged identity (F.11), the complete anatomy of *what positivity needs* — edge-band frame mass under ($\mathrm{QS}_{\exp}$)-grade non-degeneracy and $c/\log\lambda$ confinement per window — and the external welds (Suzuki, CCM). Difficulty conserved; support fully mapped; no shortcut claimed.

14.2 The assay (each technology against each face)

The two faces of (H0′)'s possible failure are zero-**location** (the dive: off-line zeros, §13–14.1 calculus) and zero-**degeneracy** (($\mathrm{QS}_{\exp}$): gap collapse). The sweep, sources read at the letter or verified by search:

Minimum-gap lower bounds — virgin ground. The entire gaps literature bounds the smallest normalized gap from *above* (small gaps exist): $\mu < 0.50895$ under RH (2026), $\mu < 1$ unconditionally (Selberg); the conjecture is $\mu = 0$. **No lower bound on the minimum gap exists or has ever been targeted.** Crucially, GUE's $\mu = 0$ is *consistent* with ($\mathrm{QS}_{\exp}$): random-matrix minimum gaps in $[0,T]$ are polynomially small — far above our exponential bar. ($\mathrm{QS}_{\exp}$) is a new, untargeted, GUE-consistent statement.

$|\zeta'(\rho)|$ moments — the weak-Gonek route. Known: $J_{-1}(T) \ge (1+o(1))\frac{3}{2\pi^3}T$ under RH+simplicity (Milinovich–Ng — the *lower* direction, unhelpful); conjectured $J_{-1} \sim \frac{3}{\pi^3}T$ (Gonek), $J_{-k} \approx T(\log T)^{(k-1)^2}$ (Gonek–Hejhal). The *upper* direction — what ($\mathrm{QS}_{\exp}$) wants — has exactly one modern result: Bui–Florea–Milinovich (BLMS 2024), **conditional on an $S(t)$-maximum conjecture and covering only a full-density subfamily of zeros.** The assay's sharpest extract:

$$(\mathrm{QS}_{\exp}) \;\Longleftarrow\; \textbf{(weak Gonek)} \quad J_{-1}(T) \ll T^A \text{ for some fixed } A, \text{ a.e. dyadic } T,$$

by Markov ($\#\{|\zeta'(\rho)| \le e^{-cT}\} \le e^{-2cT}T^A \to 0$, and a tight gap forces a small $|\zeta'|$). Gonek predicts $A = 1$; we need *any* A, on a density-one sequence of scales. Radically weaker than the conjecture; still open; the obstruction in the known technology is exceptional-zero handling (full-density subfamilies leak exceptions into every block).

Confinement rates — proportions, not exclusion. Selberg 1946: almost all zeros lie within $\psi(T)/\log T$ of the line, any $\psi \to \infty$, *unconditionally* — density one, exceptions allowed, and (H0′) tolerates no exceptions (one fixed off-line zero poisons all large windows). Levinson/Conrey/Pratt–Robles–Zaharescu–Zeindler: $c \ge 5/12$ of zeros *on* the line, unconditionally — proportions, same verdict. No exclusion technology exists at any rate.

Field positivity/screening — the usable import (Leo's placed paper). Grigutis–Turčinskas (arXiv:2509.18963, read at the letter), building on Goldštein–Grigutis: **unconditionally** (given the on-line

proportion c), for $\sigma > \frac{1}{2} + 1/\sqrt{\log t}$: $\Re \sum_{\rho \text{ on-line}} 1/(s-\rho) > (0.28-\varepsilon)c/(\sigma - \frac{1}{2})$ — an explicit positive on-line field *growing* toward the line; and their §4: under RH-failure, $\Re\, \xi'/\xi$ stays positive near the line except in small explicit neighborhoods of the off-line zeros. **This is the continuous, classical twin of our V4 shielding finding** (the on-line mass screens isolated off-line zeros; detection only within an explicit radius), with constants usable in the per-window (a)-bookkeeping. Their threshold $(\log t)^{-1/2}$ joins the log-rate hierarchy: $(\log)^{-1/2}$ field positivity, $(\log)^{-1}$ per-window insensitivity (F.14), resolution-scale detection (V4).

Verdict. Nothing unconditional touches either face; ($\mathrm{QS}_{\exp}$) is virgin ground with a named sufficient condition (weak Gonek) far below the conjectured truth; the location face is (H0′) itself, screened but not excludable by any known field technology. The open step's strength is conserved — and its support is now mapped onto two faces with named, radically-sub-conjectural sufficient conditions and explicit per-window constants.

14.3 Fetch list (the corners that could move the assay)

(1) **Bui–Florea–Milinovich, arXiv:2310.03949, full TeX** — to extract the exceptional-set structure of their conditional J_{-k} upper bounds: whether their method, aimed at exponential thresholds (vastly weaker than their polynomial targets), can be made all-zeros on a.e. dyadic blocks. (2) **arXiv:2604.05733** (small gaps, $\mu < 0.50895$ under RH, 2026) — the modern mollifier machinery at the mean-spacing scale = our F.8 constant-level race; worth reading for what its *lower*-bound byproducts are. (3) **Goldštein–Grigutis** (the predecessor of the placed paper; explicit field constants). (4) Anything on **small values of $\zeta'(\rho)$ / upper bounds for J_{-1} under RH alone** (Heap, Kirila, or adjacent) — the weak-Gonek route's nearest neighbors.

Anchors (§14). Yoshida 1992 / Suzuki Thm 3 (the (H0′) $\iff$ RH equivalence); E.20 (§22 — the erratum lands there on the referee pass); Grigutis–Turčinskas arXiv:2509.18963 + Goldštein–Grigutis (field positivity, read at the letter); Selberg 1946 (a.e. confinement; $\mu < 1$); Conrey, PRZZ ($c \geq 5/12$); Milinovich–Ng, Gonek, Hejhal, Bui–Florea–Milinovich (J_{-k}); arXiv:2604.05733 (small gaps 2026); GPD (the §13 withdrawal — the assay's first deliverable was its own correction).

15. The fetched sources at the letter: ($\mathrm{QS}_{\exp}$) calibrated exactly — named by the field, below Weak Mertens, virgin unconditionally

The fetch list landed (BFM TeX; small-gaps 2026 TeX; Goldštein–Grigutis; Durkan–Hughes–Pearce-Crump + thesis). Read at the letter; this section fixes ($\mathrm{QS}_{\exp}$)'s exact literature coordinates.

15.1 Bui–Florea–Milinovich (arXiv:2310.03949), the decisive source

(i) The families are gap-defined — the structure is ours. $\mathcal{F} = \{\gamma : |\gamma - \gamma'| \gg 1/\log T\}$ (mean-spacing separation); Theorem 1 (**RH only**): $\sum_{\gamma \in \mathcal{F}} |\zeta'(\rho)|^{-2k} \ll T^{1+\delta}$ for $2k(1+\varepsilon) \leq 1$. The $S(t)$-conjecture buys only the *enlarged family*: under RH + Farmer–Gonek–Hughes $|S(t)| \ll \sqrt{\log t \log\log t}$, separation down to $\exp(-\sqrt{\log T}/(f\sqrt{\log\log T}))$; their Remark gives the general dial: $|S(t)| \ll h(t) \Rightarrow$ separation threshold $\exp(-\log T/(f\, h(T)))$. Even the strongest conjectured $S(t)$ reaches only *subexponential* separation thresholds — and in every case the technology bounds moments over the *separated* family; **it never shows the excluded set is empty at any resolution.**

(ii) The field names our object. Their §1.2, verbatim in substance: to bound J_{-k} over *all* zeros, "simply assuming RH and the simplicity of zeros is not enough… one would need to assume **a lower bound on the size of the smallest difference between consecutive zeros**." ($\mathrm{QS}_{\exp}$)'s object, identified as the missing hypothesis by the owners of the strongest technology. The gap is real, correctly located, and not ours alone.

(iii) The Weak Mertens calibration — the sharpest upper anchor. WMC ($\int_1^X (M(x)/x)^2 dx \ll$

$\log X$) implies (Titchmarsh, Ch. 14): RH, all zeros simple, and $\sum_\rho |\rho\,\zeta'(\rho)|^{-2} < \infty$ — and convergence forces $|\zeta'(\rho)| \geq c/\gamma$ eventually, hence **polynomial** minimum gaps: **($\mathrm{QS}_{\exp}$) follows from WMC with exponential room to spare.** (BFM's own WMC-conditional full-family bounds: $J_{-k} = o(T^{k+1}(\log T)^{1-k})$ for $0 < k < 1$, $o(T^{2k})$ for $k \geq 1$.) The calibration chain, complete:

$$(\mathrm{QS}_{\exp}) \impliedby \text{any of} \quad \{\ \mathrm{RH} + \mathrm{PCC\ range}\ ;\ \ \mathrm{WMC}\ ;\ \ \text{weak Gonek } (J_{-1} \ll T^A \text{ a.e.})\ \},$$

all RH-strength or RH-conditional; **unconditionally: nothing touches it; conditionally: nothing below WMC/PCC-grade reaches exponential resolution.** Virgin ground, now with surveyed borders.

15.2 The adjacent instruments (logged as toolkit)

Small gaps 2026 (arXiv:2604.05733, read at the letter): the *resonance-correlation method* — Montgomery pair correlation synthesized with Montgomery–Odlyzko resonators through the quadratic pair-counter $N_h(t)^2 - N_h(t)$ plus an "approximator"; $\mu < 0.50895$ under RH. Two imports: the $N_h^2 - N_h$ machinery is the natural instrument for tight-pair counting in short windows (the ($\mathrm{QS}_{\exp}$) census object, if anyone ever runs it at small h); and their own remark ties the 0.5 barrier to the **Alternative Hypothesis and Siegel zeros** (Conrey–Iwaniec) — the gap statistics at half-mean-spacing touch Landau–Siegel, exactly where the program's Siegel-shadow thread (P∞-C §5.2) already sits. **Durkan–Hughes–Pearce-Crump (arXiv:2601.06292) + Durkan's thesis:** discrete second moments of mixed derivatives over zeros via the Gonek–Landau explicit formula — the same discrete-mean-value toolkit BFM build on; available for the bookkeeping stack. **Goldštein–Grigutis (arXiv:2201.08599):** the predecessor of §14.2's field-positivity import, with the explicit $A(t), B(t)$ constants. (Also flagged: the "entropy–sieve" HKO'C preprints circulating under shifting author names are treated as unverified and are not cited.)

15.3 The assay's closing verdict

($\mathrm{QS}_{\exp}$) is: **named by the field** as the missing hypothesis (BFM §1.2); **below Weak Mertens** (with exponential room), below RH+PCC, below weak-Gonek; **GUE-consistent** (random-matrix minimum gaps are polynomially small); **untargeted** — the entire gaps literature bounds the minimum from above; and **unconditionally untouched**. The location face remains (H0′) (§14.1), screened but not excludable by field-positivity technology. The program's residue therefore stands as: *(H0′) — window positivity in the limit — with its failure modes split into a location face (the per-window $c/\log\lambda$ confinement calculus, F.14) and a degeneracy face (($QS_{\exp}$), now with complete literature coordinates), plus the bookkeeping* stack.* Next mathematics, in order of provability: the bookkeeping write-out (Suzuki's continuous-kernel calculus); the weak-Gonek census at exponential thresholds via the $N_h^2 - N_h$ machinery under RH (a *conditional* but new statement, worth a strike); and the (H0′) frontier itself.

Anchors (§15). BFM arXiv:2310.03949 (Thms 1–2, §1.2's named hypothesis, the WMC section — read at the letter); Titchmarsh Ch. 14 (WMC ⇒ simplicity + convergence); arXiv:2604.05733 (resonance-correlation; $N_h^2 - N_h$; the 0.5 ↔ Siegel remark; Conrey–Iwaniec); arXiv:2601.06292 + Durkan thesis (Gonek–Landau toolkit); arXiv:2201.08599 (field constants); Kirila, Harper, Soundararajan, Bui–Florea (the moment machinery underneath BFM); §14 (the erratum and the two faces).

16. The bookkeeping strike: two lemmas proven-shaped and certified, the WMC proposition, the floor weld — and the census honestly closed (Lemmas F.15–F.16, Props F.17–F.18)

Certificate: `empirical/p_infinity/probe_pinfF_bookkeeping.py` (M1, M2 GREEN; M0's firing diagnosed through a three-round GPD chain that ended by *confirming* the lemma and depositing two numerical-methods findings).

16.1 Lemma F.15 (the tail-form bound; E.22.3 in referee form) [P**-sketch; certified**]

For h even, C^2, compactly supported, with $\int h = 0$: the Weil form on the below-window tail F_{out} of $\mathcal{E}(h)$ satisfies $|QW(F_{\text{out}})| \le C\,(1+\Delta Y)\,\|F_{\text{out}}\|_w^2$, $\|F\|_w^2 = \int |F|^2(1+|y|)\,dy$, C absolute. Proof structure: **arch+pole** via the screw-kernel calculus — Suzuki's $g_\infty''(t) = \frac{1}{2|t|} + O(1)$ turns the archimedean estimate into a three-line log-kernel bound (the $(1+|y|)$ weight is exactly the log-kernel's price); **primes** by per-shift Cauchy–Schwarz, $|(F^* * F)(\log n)| \le \sqrt{m(n)}\|F\|$ with the mass $m(n)$ below $1/(n\lambda)$ summable against $\Lambda(n)n^{-1/2}$ by Poisson decay ($\int h = 0$ kills the dual leading term); **pole** by weighted CS. **Certified (M1): $C_{\text{eff}} = 0.20, 0.24, 0.21$ across five decades of defect — log-log slope -0.00.** The tail form is norm-controlled with an absolute constant; E.22.3's "bookkeeping" label is now earned.

16.2 Lemma F.16 (cross-mixing; B2c in referee form) and the cross-ladder law [P**-sketch; law certified**]

$|QW(\mathcal{E}h_k|_{\text{win}}, g)| \le C \log\lambda\, \|F_{\text{out},k}\|\,\|g\|$ *for window g* — by E.22.1 the cross-pairing is carried entirely by the tail, then F.15's machinery bounds it. Consequence, now a *law*: the cell-to-off-$\mathcal{E}$ coupling tracks $\sqrt{\text{defect}_k}$ — **certified (M2): log-log slopes** 0.50 **(λ=2) and** 0.84 **(λ=3) across eight decades of defect.** §26's measured "0.025, Schur-stable" is upgraded: the mixing is graded by the tower, rung by rung, $\sqrt{}$-law or better.

16.3 The M0 saga (GPD record): the identity confirmed, two hypotheses refuted, two methods findings

M0 (first direct machine test of E.22.2's exact tail identity) fired at ∼0.8 relative; the diagnosis chain: (1) *h(0)-residual hypothesis* — building two-condition combos ($\int h = 0$ **and** $h(0) = 0$ exactly) changed nothing: **refuted**. (2) *Far-tail float noise* — at $y < -4$ the $\mathcal{E}$-sum has $\sim \lambda e^{|y|}$ interpolated terms cancelling to an exp-small value; the noise, pole-weighted by $e^{|y|/2}$, dominated: capping the grid collapsed the λ=2 error $0.82 \to$ **0.013** — **the identity confirmed at** 1.3% in the one case with all pieces above their noise floors. (3) *Pole-cross correction hypothesis* (that E.22.2 needs extra conditions $\int h \log w = \int wh = 0$) — **refuted by the explicit-formula bookkeeping itself**: the pole terms sit on the geometric side; the zero side vanishes termwise; the radical membership is the vanishing of the *total* form. **E.22.1/E.22.2 stand as stated** (validity-class footnote: $\mathcal{E}h$ in the explicit-formula class; $\int h = 0$ for finiteness; $h(0) = 0$ is the *Poisson-symmetry* condition of the CCM weight, not a radical condition). Deposited methods findings: **(F64-2)** pole-weighted far tails of FD-prolate images are noise minefields (Dirichlet boundary kinks ⇒ only-polynomial dual decay) — use exact Poisson evaluation or smooth-cutoff prolates; **(F64-3)** near-radical directions ($O(1)$ cancellation to exp-small) are evaluable *only* by the exact mpmath matrix — no float multiplier engine can see them. The small-λ rows' residual 37% haze is (F64-2), quantified and understood.

16.4 Proposition F.17 (WMC localizes ($\mathbf{QS}_{\text{exp}}$)) [P | WMC]

The Weak Mertens Conjecture implies $\min_{\gamma_j \le T}(\gamma_{j+1} - \gamma_j) \ge T^{-1-\varepsilon}$, hence ($QS_{\text{exp}}$) with exponential room. Proof: WMC ⇒ RH, simplicity, and $\sum_\rho |\rho\zeta'(\rho)|^{-2} < \infty$ (Titchmarsh §14); convergence forces $|\zeta'(\rho)| \ge \gamma^{-1}$ eventually; for consecutive zeros at gap g, Rolle on Ξ plus the E.17 envelope splitting $\Xi = -\text{env}\cdot Z$ gives $|\zeta'(\rho)| = |\Xi'(\gamma)|/\text{env}(\gamma) \le g \cdot \max|Z''|_{\text{loc}} \ll g\,\gamma^{\varepsilon}$ (RH-growth); combining, $g \ge \gamma^{-1-\varepsilon}$. □

16.5 Proposition F.18 (the two faces welded: ($\mathbf{QS}_{\text{exp}}$) ⟺ floor persistence) [P**-sketch; both directions certified**]

Along the windows, (QS_{exp}) holds iff the off-$\mathcal{E}$ floor satisfies $\text{floor}(\lambda) \ge e^{-c'\lambda^2}\text{poly}$. Forward: the contact law (F.14, node case; certified $g^{1.95}$) bounds the damage of any in-band pair by its gap; (QS_{exp}) keeps every contact term above the bar. Converse: a pair at gap $g \le e^{-cT}$ collapses the matching window's floor to $\sim (gL_x)^2$ below any $e^{-c'\lambda^2}$ (certified: the ×53 pair-merge of §11). **The degeneracy face is literally a**

window-Weil quantity: both faces of (H0′)'s failure — location (the dive) and degeneracy (the gap collapse) — are now statements about the same object family, the window ground and the window floor.

16.6 The census closed honestly (GPD correction)

§§11/14/15 loosely said "($\mathrm{QS}_{\mathrm{exp}}$) ⇐ RH+PCC-*range*." Corrected: the **proven** $F(\alpha)$ range cannot count sub-mean gaps at all — a Beurling–Selberg majorant of $[-\beta,\beta]$ with type-1 transform has $\int r = 2\beta + 2$: **one full mean spacing of irreducible blur** — so the census at exponential (or any sub-mean) thresholds is structurally beyond the proven range. ($\mathrm{QS}_{\mathrm{exp}}$)'s sufficient conditions are all conjectural: {PCC-at-scale, WMC, weak-Gonek}. The strike-(ii) census as originally imagined is closed; what stands is F.17–F.18 and the corrected map.

16.7 The stack after the strike

Bookkeeping* now reads: **F.15 [certified, write-out routine]; F.16 [certified law]; B2a-lower and F.11-uniformity — folded into the (H0′)-face** (they are frame statements, not bookkeeping). The ledger, in the honest final form of §14: RH $\iff$ (H0′) through the E.20-corrected amplifier, whose scaffolding is now: F.15/F.16 proven-shaped and certified, the count facet unconditional (§11), the two failure faces welded into one object family (F.18), with ($\mathrm{QS}_{\mathrm{exp}}$) carrying complete literature coordinates (§15) and a WMC-localization (F.17).

Anchors (§16). Suzuki arXiv:2606.09096 (the screw-kernel calculus — $g'' \sim 1/(2|t|)$, the lemmas' arch instrument); E.22.1–3 (P∞-E §25); Titchmarsh §14 (WMC); E.17 (envelope split); F.14 (contact law); Beurling–Selberg (the type-1 blur); GPD (M0's three-round diagnosis; two refuted hypotheses retained; F64-2/F64-3 deposited); `probe_pinfF_bookkeeping.py` + four run logs.

17. Referee capstone: the consolidated state, and the three faces of ϵ_λ — (H0′) in its most concrete costume

17.1 What this paper proved, measured, and killed (the consolidation)

Proven (unconditional): F.1 (multiplier + rank-2 pole); F.4.A (band-split floor); F.4.B (zero-evaluation annihilation — the keystone); F.6 (per-window positivity by computation); F.7 (the absorption identity: the $\mathcal{E}$-tower is the cone's sampling debt, certified to $\lambda = 5$); F.8 (edge marginality, exact); F.10 (division structure; the Hardy obstruction = RH's inner factor); F.14 (displacement calculus); F.15–F.16 (tail-form and cross-mixing bounds, constants certified flat). **Proven conditional:** F.17 (WMC ⇒ ($\mathrm{QS}_{\mathrm{exp}}$)). **Proven-shaped with certified inequalities:** F.4.C, F.11 (the merged identity), F.18 (the face-weld). **Killed by their own falsifiers and retained:** F.2 (incoherence), the diag-spike model (§8), the band-edge saturation (§7.1), the polylog-desert and no-repulsion claims (§11.2), the density race (§13.1), the fixed-rate bootstrap (§13.3 — withdrawn §14.1), the proven-range census (§16.6). **Discharged from the ledger:** (H1), by the Suzuki weld (§12). **Located:** (H0′) — window Weil positivity surviving the limit — as E.20's hidden hypothesis and the program's conserved core, with its two failure faces welded into one object family (F.18).

17.2 The (H0′) insight of the pass: ϵ_λ has three certified representations

The object of (H0′) is the window ground value ϵ_λ, and the arc has certified **three different presentations of the same number**:

$$\epsilon_\lambda \underset{\text{spectral}}{=} \min_{\|f\|=1} QW_\lambda(f) \underset{\text{geometric (E.22.2, §16.3)}}{=} QW(F_{\mathrm{out}}) \underset{\text{zero-side (F.6)}}{=} \sum_{\gamma>0} 2\,|\widehat{F}_{\mathrm{out}}(\gamma)|^2 \;+\; [\text{off-line blocks}],$$

modulo the ground-cell identification (the min is attained at the prolate cell — verified at every computed window; Suzuki's small-a continuity/asymptotic methods are the natural route to prove it). The third presentation is manifestly ≥ 0 *given confinement*; the second is computable *with no zeros mentioned.* Hence the frontier statement, in its most concrete costume yet:

(H0′-tail) *Show* $QW(F_{\text{out}}) \geq 0$ *geometrically — pole + arch − primes evaluated on one explicit function family, the prolate tails, supported in* $(0, \lambda^{-1}]$.

Two structural facts make this the right final localization. **(i) The support shrinks to the archimedean horizon**: F_{out} lives at $u \to 0$ — the cusp — and as λ grows the family concentrates there; the program's oldest localization (everything lives at the archimedean completion) reappears as the literal support of the residue. **(ii) The needed shape is published at** $\Lambda = 1$: CC-2021's Theorem 1 is *exactly* this kind of statement — a geometric Weil functional exhibited as the trace of a positive operator ($W_\infty(g \star g^*) \geq \mathrm{Tr}(\vartheta(g)\mathbf{S}\vartheta(g)^*) \geq 0$) — for the archimedean form on the prime-free window. (H0′-tail) is its semilocal version on horizon tails: the §24 "semilocal graded Sonin trace formula," now with the target family explicitly named and three certified presentations to weld any candidate proof against. Honesty: the amplifier adds no magic — (H0′)-along-a-sequence is RH-equivalent (Yoshida/Suzuki), and this paper's value is the anatomy: everything *around* (H0′) is now proven, certified, or both, and (H0′) itself is one positivity, of one explicit family, at one place, with three faces.

Anchors (§17). §§1–16 (the arc); E.22.2 + §16.3 (geometric face); F.6 (zero-side face); Suzuki Thm 3–4 (ground identification tools); CC 2021 Thm 1 (the shape of the missing proof); Yoshida (the equivalence); P∞/P0/Paper 32 (the horizon as the program's constant).

18. The (H0′-tail) strike: the horizon form's anatomy — PNT guards the edge, the conspiracy guards the interior, and the Wiener–Hopf home (Prop F.19)

Certificate: `empirical/p_infinity/probe_pinfF_h0tail.py` (T1, T3, T4 GREEN; T2's conspiracy confirmed on three of four rows, the fourth diagnosed as grid truncation; one methods deposit, F64-4). The strike attempted the §17.2 program — the semilocal CC-Thm-1 on the prolate-tail family — and its first result was a kill: **the hope "(H0′-tail) ≈ archimedean floor + small prime perturbation" is dead on the horizon.** What replaced it is a measured anatomy.

18.1 Proposition F.19 (the domain edge is the redshift line, and PNT regularizes it) [P-sketch; certified]

For horizon-supported F ($y \leq -L_x$) with L^2-mass decay $e^{2\alpha s}$ at depth s: **(i)** the pole functional $\hat{F}(i/2) = \int F e^{-y/2}$ converges iff $\alpha > \frac{1}{2}$; **(ii)** the prime sum converges iff $\sum \Lambda(n) n^{-1/2-\alpha} < \infty$ iff $\alpha > \frac{1}{2}$ — *two different terms, one critical threshold: the* $e^{y/2}$ *redshift line is the form's domain edge on the horizon.* **(iii)** As $\alpha \downarrow \frac{1}{2}$ both terms diverge like $1/(\alpha - \frac{1}{2})$ — the pole through $\hat{F}(i/2)$, the primes through $-\zeta'/\zeta(2\alpha)$ — **and the divergences cancel**: certified on pure exponential tails, pole = +15.61, −primes = −12.57, arch = −2.99, total +0.051 = zero-side +0.039 + tails at $\alpha = 0.55$ — terms of size 15 cancelling 300:1. The cancellation coefficient is the residue of ζ at $s = 1$: **the horizon form's domain-edge regularity is the prime number theorem's pole-matching, term by term. PNT guards the edge; RH guards the interior.**

18.2 The conspiracy (the interior mechanism), and the true tails' signature

Pole-dodged family ($F = (1 + (\alpha - \frac{1}{2})(y + L_x))e^{\alpha(y+L_x)}$, $\hat{F}(i/2) = 0$ by construction): at $\alpha = 1.0, 1.5, 2.5$ the pole is dead (≤ 0.25), the arch sits in its well ($-1.48, -0.74, -0.04$), and **the prime term rescues it**: −primes = $+1.37, +0.99, +0.47$, totals small-positive matching the zero side. *On the horizon, positivity off the pole is carried by arch–prime cancellation* — the seam, in geometric coordinates; any (H0′-tail) proof must encode this conspiracy, not dominate the primes. (The $\alpha = 0.75$ row fired its criterion: the dodge's

cancellation integral extends below the grid at the slowest decay — diagnosed as truncation; that row's total is still positive and zero-side-consistent.) **The true prolate tails** show the third regime: arch-*positive* ($+1.7, +2.1, +2.2$ at $\lambda = 1.3, 1.5, 2$ — their spectra live above the cone edge), prime-drag (-0.2 to -0.5), pole annihilated by the weight's two conditions: the geometric signature of ϵ_λ, with totals tracking the zero side. **Adversarial sweep:** 24 in-domain trials (oscillating envelopes, dodges, $\alpha \in [0.75, 3]$): worst total $+0.0164 \geq 0$ — the per-window positivity of the horizon form, with its empirical margin floor located (functions dodging all visible zeros approach but do not cross zero).

18.3 The Wiener–Hopf recognition, and the strike's verdict

The half-line Weil form *is* a **Wiener–Hopf operator**: QW restricted to $(-\infty, -L_x]$-supported functions is the truncation to a half-line of the convolution form whose symbol is the Weil zero-density W_∞. In that classical home: positivity of all half-line truncations $\iff$ positivity of the symbol-measure $\iff$ RH (consistent conservation); and the Krein factorization theory of Wiener–Hopf operators says positivity rides on the **spectral factorization of the symbol without inner factor** — which is precisely F.10's $\zeta(\frac{1}{2} + it) = B \cdot O$ with $B \equiv 1$. Everything welds: *(H0′-tail) in Wiener–Hopf language = the Weil symbol factorizes outer = no off-line Blaschke = RH.* **Verdict of the strike:** the semilocal CC-Thm-1 route survives only as a *factorization* statement — the positive-operator exhibit must *be* the outer factor — not as a domination statement; the horizon form's anatomy is now measured (edge = PNT; interior = conspiracy; true-tail signature; Wiener–Hopf home), the strength of the residue conserved, and no closure claimed. Methods deposit **(F64-4)**: the y-grid's spectral alias images ($t = 2\pi k/dy$) must lie outside the arch t-band — the first run's $+11 \cdot \|F\|^2$ contamination was two full alias copies of the spectrum sitting at $\theta' \approx 2.8$; the §16 bookkeeping run escaped by 142 units of t. Float64 lesson, fourth entry.

Anchors (§18). §17.2 ((H0′-tail) as posed); CC 2021 Thm 1 (the killed hope, the surviving shape); Wiener–Hopf/Krein (factorization; Hartman–Wintner-type symbol-spectrum correspondence); F.10 (inner/outer = the same question); N15 (two-potentials difference-of-convex — the conspiracy's earlier costume); PNT (the edge); GPD (the hope killed by its own derivation; T2's row diagnosed; F64-4); `probe_pinfF_h0tail.py`, `h0tail_run.log`.

19. The factorization route proper: the center-out innovation chain — RH as pointwise positivity of one computable function (Theorem F.21)

Certificate: `empirical/p_infinity/probe_pinfF_innovation_chain.py` (R1, R2 GREEN after one normalization diagnosis; two run logs). The strike took §18's verdict — *the route survives only as a factorization statement* — into Gohberg–Krein chain-factorization theory, and the chain's locality turned out to remove the limit from the open step's statement.

19.1 Theorem F.21 (the center-out chain principle) [P-structure; locality certified]

(a) Four languages, one object. For the window Weil form, the following are equivalent and their data identical: (i) positivity of QW_λ; (ii) Gohberg–Krein factorization along the spatial chain (continuous Cholesky) with positive diagonal; (iii) a de Branges/canonical-system realization on the window, the chain diagonal = the Hamiltonian density; (iv) Suzuki's $\mathcal{H}(T_a)$ construction. The chain diagonal d — the **innovation density** — is the prediction-error rate of the "Weil process."

(b) The center-out principle. Order the window by $|y|$ (center-out). Cholesky pivots depend only on the chain's past; and the Weil kernel restricted to $[-s, s]$ involves exactly the prime shifts $\log n \leq 2s$ — the primes $\leq e^{2s}$ that the $\lambda = e^s$ symbol contains — while the arch kernel and the rank-2 pole functionals are window-independent. Hence **the radius-s segment of the chain *is* the $\lambda = e^s$ window, future window growth never touches past pivots, and the entire λ-family is one chain factorization of one form.** Consequently:

$$\text{RH} \iff d(s) \geq 0 \quad \text{for every } s > 0,$$

a pointwise statement about one explicitly computable function — no limits, no tightness, no transfer. Each value $d(s)$ is a finite exact computation, immune to everything beyond radius s.

(c) Certified. All pivots positive at $\lambda = 1.3, 1.5, 1.8, 2.0$ (R1). Locality (the load-bearing structure): profiles from *different-λ builds* agree on shared radii to 4–8% once the discretizations share units — the first run's 30–40% "disagreement" was diagnosed as per-build normalization (grid spacing × basis bandwidth; the binned profiles were *parallel*, constant ratio 1.37/1.36/1.30 per pair, decomposing as $1.245 \times 1.10 \approx 1.37$ ✓); matched-bandwidth builds ($\omega_N \equiv \omega_0$) collapse it to 0.044/0.083/0.052 (R2). One universal curve.

19.2 The measured anatomy: the cascade at the growth boundary

The innovation profile $d(s)$ is smooth and decreasing, with its minimum **at the current window edge at every** λ (radii 0.235/0.378/0.559/0.665 against edges 0.262/0.405/0.588/0.693) — the seam is the growth boundary, in chain coordinates. And the exp-small ground is implemented **as a geometric cascade, not a point**: the minimum pivot falls only $1.12 \to 0.49 \to 0.045 \to 0.0059$ while the ground eigenvalue falls $2.5 \times 10^{-2} \to 1.5 \times 10^{-4} \to 4.7 \times 10^{-9} \to 1.0 \times 10^{-12}$ (the determinant accumulates the exp-smallness across many moderately-decaying edge innovations; the profile's edge decay steepens at the $d(s) \sim \exp(-c\, e^{2s})$ pace the ladder demands). En route: $\epsilon_{1.8} = 4.7 \times 10^{-9}$, a new ladder datapoint, stable across discretizations. **The probabilistic reading** (the dictionary's fifth language): the Weil process is *asymptotically deterministic* — the primes eventually determine everything, the innovation decays super-exponentially — which is P∞-E's "arithmetic stores no entropy at the horizon" made literal; **RH says determinism is approached from the positive side and never crossed.**

19.3 Status

Strength conserved ($d(s) \geq 0\ \forall s \iff$ RH). The gains: the limit is *eliminated from the statement* (pointwise in s, each value finite-exact); the martingale immunity makes every certified pivot permanent (nothing at larger radius can spoil it — the per-window verifications are now *cumulative* in a precise sense); the cascade anatomy says *how* positivity is paid (ever-smaller positive increments at the edge); and the object has a four-plus-one-language classical home. The frontier in chain coordinates: **prove the cascade stays positive** — the edge pivots are the marginal tail-form increments (the radius-s door is the F_{out} door, so §18's horizon anatomy — PNT at the edge, the conspiracy in the interior — applies to the exact cells where the cascade lives, and E.22.2's tail identity is the cascade's own increment formula). No closure claimed; the open step now sits in the Gohberg–Krein/canonical-system house with its increments named.

Anchors (§19). Gohberg–Krein (factorization along chains of projections; Volterra theory); de Branges (canonical systems; $H(t) \geq 0$ — P∞-A's equivalence, now with the chain as the constructor); Suzuki arXiv:2606.09096 ($\mathcal{H}(T_a)$; Thm 1 Friedrichs = the form being factored); Krein–Langer (screw representation = the spectral side); innovation/prediction theory (Wiener–Masani — the probabilistic face); E.22.2 (the increment formula); §18 (the edge anatomy); GPD (R2's normalization diagnosis); `probe_pinfF_innovation_chain.py`, two run logs.

20. The cascade increment resolved: the saturation wave, per-rung Cauchy–Schwarz protection, and the super-marginal decoupling of the deep ladder (Theorem F.22)

Certificate: `empirical/p_infinity/probe_pinfF_cascade_increment.py` (GREEN: S1 identity machine-exact; S2 compensation certified across eight decades; S4 positive budget remainder at every window).

20.1 Theorem F.22 (the increment decomposition and the per-rung protection) [P]

Resolve the edge pivot over the interior form's eigenbasis (eig_k, w_k):

$$d(s) \;=\; \text{bare}(e_s) \;-\; \sum_k \rho_k, \qquad \rho_k = \frac{\langle q_e, w_k\rangle^2}{\text{eig}_k}, \qquad \sigma_k := \frac{\rho_k}{\text{bare}} = \cos^2(\text{form-angle}(e_s,\ \text{mode } k)),$$

so that $d(s) > 0 \iff \sum_k \sigma_k < 1$: **the edge cell is never fully predictable from the interior.** Two structural facts are free: **(i) per-rung Cauchy–Schwarz protection** — $\rho_k \le$ bare always, in the form metric: *no single interior mode, however near-null, can overdraw the edge*; positivity can fail only by many rungs saturating jointly. **(ii)** The σ_k are exact saturation fractions summing (over the interior) to the predictable share of the edge — the open step is an energy ledger: RH $\iff$ the ledger's remainder stays positive at every radius.

20.2 The measurements: the wave, and the decoupling

Across $\lambda = 1.5, 1.8, 2.0$ (matched bandwidth; identity checked to 10^{-13}): bare edge energy λ-constant (≈ 1.91 — the local multiplier scale); even-edge pivot $d = 0.667, 0.469, 0.383$; budget $\sum\sigma = 0.651, 0.756, 0.799$, remainder $0.349, 0.244, 0.201$.

The super-marginal decoupling (S2, the strike's heart). The ground-rung correction $\rho_0 = 0.485 \to 0.0686 \to 0.0114$ *while* ϵ falls $3.6\times10^{-4} \to 1.3\times10^{-8} \to 2.4\times10^{-12}$: the edge–ground coupling^2 falls **faster** than the eigenvalue — $\rho_0 \sim \epsilon^{0.196}$, i.e. coupling $\sim \epsilon^{\beta}$ with $\beta_{\text{eff}} \approx 0.60 > \frac{1}{2}$, *above* the Cauchy–Schwarz marginal line (and inside F.16's block range 0.50–0.84). At $\lambda = 2$ the deepest mode (eig $= 2.4\times10^{-12}$) holds 0.6% of the budget, falling. **The exp-small ladder asymptotically decouples from the edge; the threat was never the depths.** *(Referee note: this claim, first made on per-rung values, is confirmed in the invariant cumulative coordinate in §22.2 — deep cumulative* 0.08–0.13 *at all mp windows, stable.)*

The saturation wave. Each rung's σ_k has a measured lifecycle: born at $O(1)$ defect, **peaking** ≈ 0.3 **at the crossover** (eig $\sim 10^{-2}$, the rung entering the bulk), then decaying $\sim \epsilon^{0.2}$ into the depth (rung 0: $\sigma = 0.254 \to 0.036 \to 0.006$ across the sweep). The budget is consumed by the *crossover region* — at $\lambda = 2$ the largest single eater is the eig $= 9.7\times10^{-3}$ rung at $\sigma = 0.32$, while rungs 0–3 (eig $\le 10^{-4}$) total 0.14. **The crossover rung is §9.3's boundary band in chain coordinates** — the seam, again, now with its energy share measured. (Parity bookkeeping: the even-edge pivot here and §19's small point-pivots reconcile as the harmonic mean of the two sectors' increments; the odd cascade carries the small one, matching §21's odd minima.)

20.3 Status: the open step as a traveling wave

$\sum\sigma(s)$ rises decelerating ($0.651, 0.756, 0.799$; increments $+0.105, +0.043$): the ledger tightens, and whether it saturates below 1 *is* the open step — stated plainly, no overclaim. The strike's gains: the per-rung lifecycle **law** (birth $\to$ crossover peak $\sigma \approx 0.3 \to \epsilon^{0.2}$-decay, with the CS bound as unconditional per-rung protection); the **relocation of the danger** from the exp-small ground (decoupled, $\beta_{\text{eff}} = 0.60$) to the crossover rung (the seam); and the final form of the residue in chain coordinates: **the saturation wave travels without its profile-sum ever reaching unity** — a statement about one measured wave profile whose every snapshot is a finite exact computation, permanent by §19's martingale immunity. Immediate next measurement: extend $\sum\sigma(s)$ to $\lambda = 2.2$–2.5 (mp-precision care: the deep rungs graze the float floor) to read the wave-sum's limit behavior against the two candidate asymptotes (saturation below 1 vs marginal creep).

Anchors (§20). §19 (the chain, the immunity); F.16 (the cross-ladder law — β re-measured at the edge cell: 0.60); §9.3 (the boundary band = the crossover); Cauchy–Schwarz in indefinite-free form metrics; §21 of P∞-E (odd-sector minima — the parity reconciliation); GPD; `probe_pinfF_cascade_increment.py`, `cascade_increment_run.log`.

21. The wave limit measured: the remainder law $R \approx Ce^{-0.34\lambda^2}$, the 29% eating fraction, and the open step as wave-stationarity

Certificate: `empirical/p_infinity/probe_pinfF_wave_limit.py` (mp, dps = 50; W1 validation exact; the sweep to $\lambda = 2.5$ resolving eigenvalues to 1.55×10^{-23}). **The strike's first result is a GPD self-correction: the registered W2 dichotomy — "saturation below 1 vs creep to 1" — was mis-**

designed. Under RH-*true*, $\sum\sigma \to 1$ is *forced*: §19.2's own dictionary (the Weil process is asymptotically deterministic) requires the determinant to die and the increments to vanish. The discriminator is not the limit of $\sum\sigma$ but the **law of the remainder** $R = 1 - \sum\sigma$: lawful positive decay (RH-shaped) versus a zero crossing (violation). The rule fired "creep-favored" and thereby exposed its own framing error — recorded, retained.

21.1 The measurement (mp dps = 50; deep rungs by mp inverse iteration; $\sum\sigma$ by mp-LU, deflation-immune)

$$\sum\sigma \;=\; 0.6509,\ 0.7563,\ 0.7992,\ 0.8600,\ 0.8867,\ 0.9125 \qquad (\lambda = 1.5, 1.8, 2.0, 2.2, 2.35, 2.5),$$

validated against the float run at $\lambda = 2$ (0.7992 exactly). The remainder: $R = 0.3491, 0.2437, 0.2008, 0.1400, 0.1133, 0.08$
and per-interval $d(\ln R)/d(\lambda^2) = -0.36, -0.26, -0.43, -0.31, -0.35$:

$$\boxed{R(\lambda) \;\approx\; C\,e^{-c\lambda^2}, \qquad c = 0.34 \pm 0.06}$$

— **the new architectural constant** (refined global fit on the mp truths: $c = 0.342$). The law reads: $\approx 29\%$ **of the remaining budget is eaten per unit** λ^2 ($R_{n+1} \approx 0.71\,R_n$ per unit). *(Referee note, post-§22: the first-draft identification of* 0.29 *with the per-rung crossover peak* $\sigma_{\text{peak}} \approx 0.3$ *rested on per-rung values and a migrating ladder boundary — retracted in §22.1(iii); the certified statement is the law itself, with the branch-resolved per-arrival decomposition deferred to parity-total measurements, §22.3.)* The mechanism direction stands: *stationary wave ⇒ constant eating fraction ⇒ geometric remainder ⇒ positive forever;* $c = 0.342$ is the §19.2 cascade pace $d(s) \sim \exp(-c\,e^{2s})$ with its constant measured.

21.2 Supporting data, and flags

σ_0-decoupling continues: $5.9 \times 10^{-3} \to 1.0 \times 10^{-3} \to 3.4 \times 10^{-4} \to 1.2 \times 10^{-4}$ — the exp-small depths are irrelevant, now confirmed to eig $= 1.55 \times 10^{-23}$. **New ladder datapoints:** $\epsilon(2.2) = 2.0 \times 10^{-16}$, $\epsilon(2.35) = 8.3 \times 10^{-20}$, $\epsilon(2.5) = 1.55 \times 10^{-23}$ — the Fuchs *shape* exact across what is now a **21-decade run** ($10^{-2} \to 10^{-23}$), constant prefactor ratio with a mild subleading drift (logged; the F3 ratios 3.3, 7.7 continue ~ 22 at 2.5 — a λ-power beyond the quoted λ^9, flagged for the Fuchs write-out). Honest flag: the mp deflation scrambled individual deep-rung σ entries (a duplicated ground at $\lambda = 2.5$); $\sum\sigma$ comes from the LU solve and is immune; per-rung deep-σ values are indicative only.

21.3 The open step, final form of the arc

$$\text{RH} \iff d(s) > 0\ \forall s \iff R(s) > 0\ \forall s \impliedby \textbf{wave-stationarity}\ \text{(the lifecycle profile stays sub-unit)},$$

with the last implication the program's sharpest sufficient condition: a *self-similarity* statement — each rung's saturation lifecycle is the same curve in its own scale (measured at six windows: birth → peak ≈ 0.3 at crossover → $\epsilon^{0.2}$ decay; eating fraction 0.29 per rung, constant). What would prove it: a renormalization argument for the rung lifecycle — the chain's increment structure (§20) maps rung n's lifecycle to rung $n{+}1$'s under the scaling $\lambda^2 \to \lambda^2 + 1$; stationarity of that map's fixed profile is exactly the kind of statement RG/self-similarity machinery addresses, and every ingredient of the map (tail identities, cross-ladder law, crossover geometry) now carries certified bounds. Strength conserved — stationarity-forever is the open step — but the support has never been smaller: **one traveling profile, one fixed point, one fraction that must stay below one.**

Anchors (§21). §§19–20 (the chain, the ledger); F.7 (rung arrival rate); §19.2 (determinism — the corrected dichotomy); Fuchs (the 21-decade shape); GPD (the W2 rule's framing error, caught by its own firing); mp inverse iteration/LU; `probe_pinfF_wave_limit.py`, `wave_limit_run.log`.

22. The lifecycle map in invariant coordinates: the cumulative ledger, the settled anomaly, and the two-branch structure

Certificates: `probe_pinfF_lifecycle_rg.py` (two runs; C1, C1′ fired), `probe_pinfF_lifecycle_parity.py` (D1 GREEN), `probe_pinfF_cumulative.py` (E1, E2 GREEN; E3, E4 fired — the invariance audit); clouds archived in `lifecycle_clouds.npz`. **This section was rewritten before consolidation after an internal-consistency catch (L.): the first draft's per-rung claims used quantities §21's own flag had declared non-invariant.**

22.1 The kills (GPD; all retained)

(i) Single master curve $\sigma = \Psi(\text{eig})$: DEAD — cross-window dispersion 0.64–0.79 at fixed eigenvalue, **normalization-independent** ($\div R(\lambda)$: $0.792 \to 0.789$) — never amplitude scaling. **(ii) Float-corruption diagnosis:** the $\lambda \geq 2.35$ float deep bins disagree with the dps-50 truth ($\sum\sigma(2.5)$: 0.8834 vs 0.9125; non-monotone R) — float eigh on 10^{20}-conditioned matrices pollutes near-null eigenvectors (F64-3 extended). **(iii) The first draft's "budget closes to 1%" claim: RETRACTED** — it was built on per-rung bin-means (non-invariant under near-degenerate rotations — the same defect as the deflation-scrambled entries) and on a fixed-eigenvalue ladder boundary that the crossover *migrates across* (between $\lambda = 1.8$ and 2.0 the ladder-cumulative grew 0.130 while total $\sum\sigma$ grew 0.043: boundary migration, not eating). The invariance audit (E4) fired exactly there.

22.2 The invariant coordinate: the cumulative ledger, and the settled anomaly

The basis-free object is the **cumulative saturation** $s(\xi) = q^\top f_\xi(K_{\text{int}})\, q/\text{bare}$, $f_\xi(x) = x^{-1}\mathbf{1}_{x\leq\xi}$ — a spectral function, immune to within-cluster rotations and deflation scrambling; the deep part is computable *exactly* as [mp total via LU] − [float upper part] (modes above 10^{-6} mix with the deep block only at $\sim 10^{-10}$). Two settlements: **(E1)** the dps-50 log's warning-shot entry (eig $= 2.33\times10^{-10}$, $\sigma = 0.31$ at $\lambda = 2$ — matching no trustworthy rung, suspiciously equal to the crossover's 0.32) is **certified a deflation artifact**: the invariant deep cumulative at $\lambda = 2$ is $s(\leq 10^{-6}) = 0.127$, in the registered band and matching the float table's 0.139; the mechanism is identified (imperfect deflation ⇒ mixed vector ⇒ Rayleigh quotient *between* rungs with the shallower component's coupling — inflated ratio). **(E2)** §20's danger-relocation claim **stands in invariant form**: the depths hold $0.127, 0.090, 0.082, 0.113$ of the budget at $\lambda = 2.0, 2.2, 2.35, 2.5$ — $\sim 10\%$, stable, no growth.

The ledger's structure (E3's instructive firing): the cumulative curve is a *mixed object* — a **λ-stationary bulk** (fixed in absolute eigenvalue) plus a **migrating ladder** (Fuchs pace); a single global shift cannot collapse both, and the small fitted shifts (slope 0.53 vs 4π) measure precisely that mixture. The established self-similarity is the *ladder's*, parity-resolved: **two interleaved towers** (even mod-4 and its odd twin; the edge is parity-mixed via the DFT grid's i=0 fixed point; at fixed eigenvalue the two parities sit at different ages), with within-branch dispersion 0.107 **(even) and** 0.019 **(odd)** versus 0.79 mixed (D1, four clean windows; populated-bin comparisons, parity-splitting halves occupancy).

22.3 The honest budget

The certified closure is the **remainder law itself**: $R(\lambda) \approx Ce^{-c\lambda^2}$ with $c = 0.342$ (mp-truth global fit; the float tail biases low). The branch structure that survives invariantly: the even tower dominates the ladder ledger at every window (even cumulative $\gg$ odd, $\sim$0.25–0.45 vs 0.05–0.08); the *per-arrival* branch decomposition of the eating requires parity-resolved **totals** across windows (differences of parity-cumulatives, free of the migrating boundary) — flagged as the next measurement, **not claimed**. The flow check on safe pairs stands as before (median +0.094, n=4, small).

22.4 The proof target in seam-relative coordinates

The open step: **the migrating ladder-ledger's increment per arrival stays bounded below the remainder** — equivalently the two parity-branch profiles (D1-established at 2–11%) remain stationary and sub-unit under the one-step transfer $\lambda^2 \to \lambda^2 + 1$, with the remainder law $c = 0.342$ as the integral statement. Everything feeding the map carries certified bounds (E.22.2/F.15 tails, F.16 cross-ladder, §20 crossover geometry); $\sum \sigma < 1 \ \forall s \iff R > 0 \iff d > 0 \iff$ RH, unchanged in strength. The fixed profile lives where the Cauchy–Schwarz protection and the remainder law always lived: **in the cumulative ledger.**

Anchors (§22). §§19–21 (chain, ledger, remainder law); F.7 (arrival rates); §21 of P∞-E (the odd tower); the internal-consistency catch (L., pre-consolidation — the per-rung/cumulative invariance distinction); GPD (four kills retained: C1, C1′, the 1%-closure, E3-as-instrument); `probe_pinfF_lifecycle_rg.py`, `probe_pinfF_lifecycle_parity.py`, `probe_pinfF_cumulative.py`, `lifecycle_clouds.npz`.